Algebra: Chapter 0 - Self Study

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Abstract

This text contains my notes on Algebra: Chapter 0, by Paolo Aluffi. I read this book independently, starting over winter break 2024. My notes largely paraphrase the text, and include few examples. I complete some examples as I see fit.. I read chapters 1, 2, and 3 over the course of winter break, and read chapters 4-9 over the spring semester. I did not read them in order; I read chapter 6, 8, and 9 before most of chapters 4 and 5.

Contents

Chapter 1

Homological Algebra

Homological algebra can be motivated by the study of the Tor and Ext functors. We will study objects such as *derived functors*, which are important in constructions such as sheaf cohomology. We will work in abelian categories, which are categories with good notions of kernels and cokernels.

1.1 (Un)necssary Categorical Preliminaries

1.1.1 Undesirable Features of Otherwise Reasonable Categories

We have encountered categories that have annoying features such as

- A category may not have initial or final objects, and they may not be the same object.
- Products and coproducts may not exist (for example, Fld).
- Kernels and cokernels may not exist for all morphisms.
- Even if kernels and cokernels exist, monomorphisms need not be kernels, and epimorphisms need not be cokernels.
- A morphism may be mono and epi without being an isomorphism.

Homological algebra deals almost entirely with kernels and cokernels, as it deals with complexes and exactness. We have already used this language in the category R-Mod, in which none of these annoying properties were true. We can generalize these notions.

1.1.2 Additive Categories

Definition 1.1: Additive

A category A is *additive* if it has a zero-object (an object that is both initial and final), finite products and coproducts, and $\mathrm{Hom}_{\mathsf{A}}(A, B)$ has an abelian group structure for all $A, B \in \mathsf{Obj}(\mathsf{A})$ such that the composition maps are bilinear. A functor $\mathscr{F} : \mathsf{A} \to \mathsf{B}$ between additive categories is additive if it preserves the abelian group structure of the Hom-sets.

Since Hom-sets are abelian groups, they have identity objects. We refer to these as zero-morphisms, denoted by 0. We can also define addition and subtraction of morphisms, and we can define morphism equality: $\alpha = \beta$ if and only if $\alpha - \beta = 0$. Using this language, we can define kernels and cokernels. In our previous definitions, we defined these as an object of our category, but the information in a kernel or cokernel is really carried by the injections and projections, respectively, that they carry. We thus give the definition for an arbitrary category as follows.

Definition 1.2: Kernel/Cokernel

Let $\phi : A \to B$ be a morphism in an additive category A. A morphism $\iota : K \to A$ is a *kernel* of ϕ if $\phi \circ \iota = 0$ and for all morphisms $\zeta : Z \to A$ such that $\phi \circ \zeta = 0$, there is a unique $\widetilde{\zeta}$ such that

$$\begin{tikzcd}
Z \arrow[r, "\zeta"'] \arrow[rr, bend left, "0"] \arrow[dr, dashed, "\widetilde{\zeta}"'] & A \arrow[r, "\phi"'] & B \\
& K \arrow[u, "\iota"] &
\end{tikzcd}$$

commutes. A morphism $\pi : B \to C$ is a *cokernel* of ϕ if $\pi \circ \phi = 0$ and for all morphisms $\beta : B \to Z$, there exists a unique $\widetilde{\beta} : C \to Z$ such that

$$\begin{tikzcd}
& C \arrow[dr, dashed, "\widetilde{\beta}"] & \\
A \arrow[r, "\phi"] \arrow[rr, bend right, "0"'] & B \arrow[u, "\pi"] \arrow[r, "\beta"] & Z
\end{tikzcd}$$

commutes.

More succinctly, if $\phi \circ \zeta = 0$, then ζ factors uniquely through $\ker \phi$, and if $\beta \circ \phi = 0$, then β factors uniquely through $\operatorname{coker} \phi$.

It is also prudent to recall the definitions of monomorphism and epimorphism from chapter 1, all that time ago.

Definition 1.3: Mono/Epi

A morphism $\phi A \to B$ is a monomorphism if for all $\zeta_1, \zeta_2 : Z \to A$, if

$$\begin{tikzcd}
Z \arrow[r, shift left, "\zeta_1"] \arrow[r, shift right, "\zeta_2"'] & A \arrow[r, "\phi"] & B
\end{tikzcd}$$

commutes, then $\zeta_1 = \zeta_2$. It is an epimorphism if for all $\beta_1 \beta_2 : B \to Z$, if

$$\begin{tikzcd}
A \arrow[r, "\phi"] & B \arrow[r, shift left, "\beta_1"] \arrow[r, shift right, "\beta_2"'] & Z
\end{tikzcd}$$

commutes, then $\beta_1 = \beta_2$.

We can simplify this somewhat in abelian categories.

Lemma 1.1

A morphism $\phi : A \to B$ in an additive category is a monomorphism if and only if for all $\zeta : Z \to A$,

$$\phi \circ \zeta = 0 \implies \zeta = 0.$$

ϕ is an epimorphism if for all $\beta : B \to Z$,

$$\beta \circ \phi = 0 \implies \beta = 0.$$

Proof. We will prove this for monomorphisms, since the proof for epimorphisms is equivalent. First, let ϕ be mono. Since composition maps are bilinear, $\phi \circ -$ is linear, and thus a group homomorphism. Thus, $\phi \circ 0 = 0$, since identities map to identities. We then have $\phi \circ \zeta = \phi \circ 0$, and thus $\zeta = 0$.

Now for the converse, let $\phi \circ \zeta = 0$ imply $\zeta = 0$ for all $\zeta : Z \to A$. Let $\alpha, \beta : Z \to A$, and suppose that $\phi \circ \alpha = \phi \circ \beta$. Then $\phi \circ \alpha - \phi \circ \beta = 0$, and since $\phi \circ -$ is bilinear, we have

$$\phi \circ (\alpha - \beta) = \phi \circ \alpha - \phi \circ \beta.$$

It follows that $\alpha - \beta = 0$, and thus $\alpha = \beta$. ■

Moreover, we can show that kernels are monomorphisms, and cokernels are epimorphisms.

Lemma 1.2

In an additive category, kernels are monomorphisms and cokernels are epimorphisms.

Proof. I will prove the statement for monomorphisms, since the statement for epimorphisms is the same as the statement for monomorphisms in the opposite category. This category is naturally an additive category as well, so it suffices to show only one side of the proof.

Let $\phi : A \to B$ be a morphism in an additive category, and let $\ker \phi : K \to B$ exist. If $\gamma : Z \to A$ is a morphism such that $\ker \phi \circ \gamma = 0$, then $\phi \circ (\ker \phi \circ \gamma) = \phi \circ 0 = 0$, and thus γ factors uniquely through $\ker \phi$:

$$\begin{tikzcd}
Z \arrow[r, "\ker \phi \circ \gamma = 0"] \arrow[dr, bend left, "\gamma"] \arrow[dr, dashed, bend right] & A \arrow[r, "\phi"] & B \\
& K \arrow[u, "\ker \phi"'] &
\end{tikzcd}$$

We thus have $\gamma \circ \ker \phi = 0$, and we know that $0 \circ \ker \phi = 0$. Therefore, $\gamma = 0$ by uniqueness, and ϕ is a monomorphism by the previous lemma. ■

Kernels and cokernels may not exist. However, if a morphism has kernels and cokernels, then we can recover expected information from them. We can also guarentee the existance of some of them.

Lemma 1.3

Let $\phi : A \to B$ be a morphism in an additive category. Then ϕ is a monomorphism if and only if $0 \to A$ is its kernel, and ϕ is an epimorphism if and only if $B \to 0$ is its cokernel.

Proof. I will prove the statement for cokernels this time. Let $\phi : A \to B$ be an epimorphism, and let $\beta : B \to Z$ such that $\beta \circ \phi = 0$. It follows that $\beta = 0$ by lemma 1.1, and thus

$$\begin{tikzcd}
A \arrow[r, "\phi"'] \arrow[rr, bend left, "0"] & B \arrow[r, "\beta"'] \arrow[d, "0"] & Z \\
& 0 \arrow[ur, dashed, "0"'] &
\end{tikzcd}$$

commutes for all β. Moreover, since $0 : 0 \to Z$ is unique since 0 is initial, we have that $0 : B \to 0$ is a cokernel of ϕ.

For the converse, suppose that $\operatorname{coker}\phi = 0 : B \to 0$. It follows that for all $\beta : B \to Z$ such that $\beta \circ \phi = 0$, the diagram

$$\begin{tikzcd}
A \arrow[r, "\phi"'] \arrow[rr, bend left, "0"] & B \arrow[r, "\beta"'] \arrow[d, "0"] & C \\
& 0 \arrow[ur, dashed, "0"'] &
\end{tikzcd}$$

commutes, and thus $\beta = 0 \circ 0 = 0$. By lemma 1.1, ϕ is an epimorphism. ■

We can thus begin discussing exactness of sequences in the context of these modules, such as

$$0 \longrightarrow A \longrightarrow B, \qquad A \longrightarrow B \longrightarrow 0$$

for mono and epi, respectively. However, we cannot generalize exactness of sequences unless kernels and cokernels exist in general.

We generally denote monomorphisms by $\rightarrowtail$ and epimorphisms by $\twoheadrightarrow$.

1.1.3 Abelian Categories

Abelian categories are obtained by simply taking the features we desire from additive categories, and explicity demanding them.

Definition 1.4: Abelian Category

An additive category A is *abelian* if kernels and cokernels exist for all morphisms in A, every monomorphism is a kernel, and every epimorphism is a cokernel.

We call these categories abelian because the most basic prototype is Ab, however there is nothing particularly commutative about these categories. We have also seen that R-Mod is an abelian category. Moreover, by lemma 1.2, we have found that in abelian categories, a morphism is mono if and only if it is a kernel, and it is epi if and only if it is a cokernel. We can thus obtain an intuitive understanding of monomorphisms $A \rightarrowtail B$ as identifying A as a subobject of B, and conversely epimorphisms $A \twoheadrightarrow B$ as quotients; if $\phi : A \to B$ is a monomorphism, then B/A can denote the target of $\operatorname{coker}\phi$.

Lemma 1.4

In an abelian category A, every kernel is the kernel of its cokernel, and every cokernel is the cokernel of its kernel.

Proof. First, let $\phi : A \to B$ be the kernel of a morphism $f : B \to Z$. Since A is abelian, there exists a cokernel $\pi : Z \to K$ of ϕ, and since the composition $f \circ \phi : A \to B \to Z$ is 0, the map $f : B \to Z$ factors uniquely through $\operatorname{coker}\phi$:

$$\begin{tikzcd}
& K \arrow[dr, dashed] & \\
A \arrow[r, "\phi"] \arrow[rr, bend right, "0"'] & B \arrow[u, "\pi"] \arrow[r, "f"] & Z
\end{tikzcd}$$

Now let $g : C \to A$ such that $g \circ \pi = 0$. Then

$$\begin{tikzcd}
 & & K \arrow[dr, dashed] & \\
A \arrow[r, "\phi"] & B \arrow[ur, "\pi"] \arrow[rr, "f"] & & Z \\
 & C \arrow[u, "g"] & &
\end{tikzcd}$$

commutes, and thus $f \circ g = 0$:

$$\begin{tikzcd}
 & & K \arrow[dr, dashed] & \\
A \arrow[r, "\phi"] & B \arrow[ur, "\pi"] \arrow[rr, "f"] & & Z \\
 & C \arrow[u, "g"] \arrow[urr, dashed, bend right, "0"] & &
\end{tikzcd}$$

It follows that g factors uniquely through $\ker f = \phi$, but we also have that g factors uniquely through $\ker \pi$ by definition. Therefore, $\ker f$ satisfies the universal property for $\ker \pi$, and thus $\ker \operatorname{coker} \ker f = \ker f$. ■

By these lemmas, we can recharacterize the definition of an abelian category A as simply an additive category with kernels and cokernels, such that

- If $\phi : A \to B$ has $\ker \phi = 0$, then $\phi = \ker \operatorname{coker} \phi$;
- If $\psi : B \to C$ has $\operatorname{coker} \psi = 0$, then $\psi = \operatorname{coker} \ker \psi$.

This definition immediately characterizes when a short sequence

$$0 \longrightarrow A \xrightarrow{\phi} B \xrightarrow{\psi} C \longrightarrow 0$$

is exact. We must have $\psi = \operatorname{coker} \phi$, and $\ker \psi = \phi$. Either one of these is equivalent, since ϕ must be mono and ψ must be epi, and thus if $\psi = \operatorname{coker} \phi$, then

$$\ker \psi = \ker \operatorname{coker} \phi = \phi,$$

and vice versa.

We can also consider exact sequences

$$0 \longrightarrow A \overset{\phi}{\rightarrowtail\!\!\twoheadrightarrow} B \longrightarrow 0.$$

For this to be exact, we must have $\ker \phi = 0$ and $\operatorname{coker} \phi = 0$, and thus ϕ is both mono and epi. In abelian categories, this implies ϕ is an isomorphism.

Lemma 1.5

Let $\phi : A \to B$ be a monomorphism and an epimorphism. Then ϕ is an isomorphism.

Proof. We have

- $\ker \phi$ is $0 \to A$;
- $\operatorname{coker} \phi$ is $B \to 0$;
- $\operatorname{coker} 0 \to A$ is ϕ;
- $\ker B \to 0$ is ϕ

by lemmas 1.3 and 1.4. We then have

$$\begin{tikzcd} & & B \arrow[d, "1_B"] & \\ 0 \arrow[r] & A \arrow[r, "\phi"] & B \arrow[r] & 0 \end{tikzcd}$$

commutes, since clearly $B \to B \to 0$ is 0. It thus factors uniquely through $\ker B \to 0$:

$$\begin{tikzcd} & & B \arrow[d, "1_B"] \arrow[dl, dashed, "\psi"] & \\ 0 \arrow[r] & A \arrow[r, "\phi"'] & B \arrow[r] & 0 \end{tikzcd}$$

Since this diagram commutes, $\phi \circ \psi = 1_B$, and thus ϕ has a right-inverse.

Now for the right inverse, consider the diagram

$$\begin{tikzcd} 0 \arrow[r] & A \arrow[r, "\phi"] \arrow[d, "1_A"] & B \arrow[r] & 0 \\ & A & & \end{tikzcd}$$

Since $1_A \circ 0 = 0$ clearly, 1_A factors uniquely through $\operatorname{coker} 0 \to A = \phi$:

$$\begin{tikzcd} 0 \arrow[r] & A \arrow[r, "\phi"] \arrow[d, "1_A"] & B \arrow[r] \arrow[dl, dashed, "\eta"] & 0 \\ & A & & \end{tikzcd}$$

Since this diagram commutes, we have $\eta \circ \phi = 1_A$, and thus ϕ has a left-inverse. By a proof done in chapter 1, it follows that this inverse is a unique two-sided inverse $\eta = \psi$, and thus ϕ is an isomorphism. ■

1.1.4 Products, Coproducts, and Direct Sums

In an abelian category A, we will denote the product of A and B by $A \times B$, and we will denote the coproduct $A \amalg B$. These exist, since A is additive, and thus contains finite products and coproducts by definition. We will see later that we can substitute the notation $A \oplus B$ for both of these.

The presence of products, coproducts, kernels, and cokernels allows us to consider fairly many important constructions. For example, pullbacks (also fibered products) exist in abelian categories. If

$$\begin{tikzcd} & B \arrow[d, "\psi"] \\ A \arrow[r, "\phi"] & C \end{tikzcd}$$

is a diagram in A, then we can define the fibered product of A and B over C as an object $A \times_C B$ that is final with respect to commutative diagrams

$$\begin{tikzcd} A \times_C B \arrow[r, "\psi'"] \arrow[d, "\phi'"'] & B \arrow[d, "\psi"] \\ A \arrow[r, "\phi"'] & C \end{tikzcd}$$

That is,

$$\begin{tikzcd} P \arrow[dr, dashed] \arrow[drr, bend left] \arrow[ddr, bend right] & & \\ & A \times_C B \arrow[r] \arrow[d] & B \arrow[d] \\ & A \arrow[r] & C \end{tikzcd}$$

The pullback may be constructed explicitly as follows. Note that since $A \times B$ exists, there exist canonical projections $\rho_A, \rho_B : A \times B \to A, B$ respectively. We thus have compositions

$$A \times B \xrightarrow{\rho_A} A \xrightarrow{\phi} C,$$

$$A \times B \xrightarrow{\rho_B} B \xrightarrow{\psi} C.$$

It is shown in the exercises the kernel of the difference of these morphisms can construct a pullback. Moreover, fibered coproducts/pullouts can also be constructed as the dual of this statement.

Such constructions in abelian categories are heavily simplified by the following fact, which is also true in additive categories.

Proposition 1.1

Let A be an additive category. Then finite products and coproducts coencide; that is, $A \times B \cong A \amalg B$.

Proof. By the universal property of coproducts, and since zero-morphisms exist in additive categories, the diagram

$$\begin{tikzcd} A \arrow[dr, "i_A"] \arrow[drr, bend left, "\mathrm{id}"] & & \\ & A \amalg B \arrow[r, dashed, "\pi_A"] & A \\ B \arrow[ur, "i_B"] \arrow[urr, bend right, "0"'] & & \end{tikzcd}$$

commutes. By an equivalent construction into B, we have morphisms

$$\begin{tikzcd}
A \arrow[rr, "1_A"] \arrow[dr, "i_A"] & & A \\
& A \amalg B \arrow[ur, "\pi_A"] \arrow[dr, "\pi_B"] & \\
B \arrow[rr, "1_B"] \arrow[ur, "i_B"] & & B
\end{tikzcd}$$

By the universal property of products, there exists a unique morphism $(\pi_A \times \pi_B) : A \amalg B \to A \times B$ such that

$$\begin{tikzcd}
& & A \\
A \amalg B \arrow[r, dashed, "(\pi_A \times \pi_B)"] \arrow[urr, bend left, "\pi_A"] \arrow[drr, bend right, "\pi_B"'] & A \times B \arrow[ur] \arrow[dr] & \\
& & B
\end{tikzcd}$$

commutes. I will show that $A \amalg B$ satisfies the universal property of products, together with π_A, π_B.

Let $Z \to A, B$ be morphisms. Then the diagram

$$\begin{tikzcd}
& A \arrow[rr, "1_A"] \arrow[dr, "i_A"] & & A \\
Z \arrow[ur] \arrow[dr] & & A \amalg B \arrow[ur, "\pi_A"] \arrow[dr, "\pi_B"] & \\
& B \arrow[rr, "1_B"] \arrow[ur, "i_B"] & & B
\end{tikzcd}$$

commutes, due to the identity morphisms. I can't figure out how to conclude this purely arrow-theoretically without appealing to abelian categores, but it seems fairly obvious why it's true. I'll work on it later. ■

We thus can adobt the notation $A \oplus B$ for both $A \times B$ and $A \amalg B$ in any abelian category, and we call it the direct sum of A and B. It comes with natural morphisms

$$A, B \rightarrowtail A \oplus B \twoheadrightarrow A, B$$

satisfying all properties we would expect.

1.1.5 Images, Canonical Decomposition of Morphisms

Arrow-theoretical approaches allow us to define a canonical decomposition of morphisms in any abelian category. That is, we should be able to decompose any morphism into a monomorphism, an epimorphism, and a unique isomorphism. This result requires multiple lemmas, and will induce an intuitive definition of image. The statement we will prove is that any morphism $\phi : A \to B$ decomposes as

$$\begin{tikzcd}
A \arrow[r, two heads, "\operatorname{coker}\ker\phi"'] \arrow[rrr, bend left, "\phi"] & C \arrow[r, dashed, tail, two heads] & K \arrow[r, tail, "\ker\operatorname{coker}\phi"'] & B.
\end{tikzcd}$$

We would assume that $\ker \operatorname{coker} \phi$ should play the role of $\operatorname{im} \phi$. We can formalize this notion somewhat. Note that in the context of sets, any subset B' such that $\operatorname{im} \phi \subseteq B' \subseteq B$ has the property that the mapping $\phi : A \to B'$ factors uniquely through $\operatorname{im} \phi$. By replacing "subobject" with "monomorphism", we can thus see $\operatorname{im} \phi$ as initial with respect to being a monomorphism $\_ \to B$ through which ϕ factors.

Lemma 1.6

Let $\phi : A \to B$ be a morphism in an abelian category, and let $\iota : K \to B$ be $\ker \operatorname{coker} \phi$. Then

- ι is a monomorphism;
- ϕ factors through ι;
- ι is initial with respect to these properties.

Proof. Since ι is a kernel in an abelian category, it is a monomorphism. Since the composition

$$A \xrightarrow{\phi} B \xrightarrow{\operatorname{coker} \phi} L$$

is 0 by definition of cokernel, ϕ factors uniquely through $\ker \operatorname{coker} \phi = \iota$:

$$\begin{tikzcd} A \arrow[r] \arrow[rr, bend left, "\phi"] & K \arrow[r, "\iota"] & B \end{tikzcd}$$

This shows the first two properties. Now suppose $\lambda : L \to B$ is mono and factors through ϕ. We want to show ι factors through λ:

$$\begin{tikzcd} & L \arrow[dr, "\lambda"] & \\ A \arrow[ur] \arrow[r] \arrow[rr, bend right, "\phi"'] & K \arrow[u] \arrow[r, "\iota"] & B \end{tikzcd}$$

Since $\operatorname{coker} \lambda \circ \lambda = 0$, we have $\operatorname{coker} \lambda \circ \phi = 0$, and thus $\operatorname{coker} \lambda$ factors uniquely through $\operatorname{coker} \phi$:

$$\begin{tikzcd} & L \arrow[dr, tail, "\lambda"] & & \\ A \arrow[ur] \arrow[r] \arrow[rr, bend right, "\phi"'] & K \arrow[r, tail, "\iota"] & B \arrow[r, "\operatorname{coker} \phi"] \arrow[dr, "\operatorname{coker} \lambda"'] & \operatorname{Cok} \phi \arrow[d, dashed] \\ & & & \operatorname{Cok} \lambda \end{tikzcd}$$

It follows fairly obviously that $\operatorname{coker} \lambda \circ \iota = 0$, and thus ι factors uniquely through $\ker \operatorname{coker} \lambda$. Since λ is mono and A is abelian, $\ker \operatorname{coker} \lambda = \lambda$. ■

This lemma motivates the following definition.

Definition 1.5: Image/Coimage

Let $\phi : A \to B$ be a morphism in an abelian category. The image $\operatorname{im} \phi$ is defined as $\ker \operatorname{coker} \phi$, and the coimage $\operatorname{coim} \phi$ is defined as $\operatorname{coker} \ker \phi$.

To summarize, for any morphism $\phi : A \to B$, we have a decomposition

$$\begin{tikzcd}
 & K \arrow[dr, tail, bend left, "\operatorname{im}\phi"] & \\
A \arrow[ur, bend left, "\overline{\phi}"] \arrow[rr, "\phi"] \arrow[dr, two heads, bend right, "\operatorname{coim}\phi"'] & & B \\
 & C \arrow[ur, bend right, "\underline{\phi}"'] &
\end{tikzcd}$$

Our intuition would suggest that $\overline{\phi}$ is epi and $\underline{\phi}$ is mono. This is indeed the case.

Lemma 1.7

Let $\phi : A \to B$ be a morphism in an abelian category, and let $\operatorname{im} : K \to B$, $\operatorname{coim} : A \to C$. Then the induced morphisms $A \to K$ and $C \to B$ are epimorphisms and monomorphisms, respectively.

Proof. To show $\underline{\phi} : C \to B$ is a monomorphism, consider its coimage $C \to C'$:

$$\begin{tikzcd}
 & & C' \arrow[ddr, "\underline{\underline{\phi}}"] & \\
 & C \arrow[ur, two heads, "\operatorname{coim}\underline{\phi}"] \arrow[drr, "\underline{\phi}"] & & \\
A \arrow[ur, two heads, "\operatorname{coim}\phi"] \arrow[rrr, "\phi"] & & & B
\end{tikzcd}$$

We thus have a factorization

$$\begin{tikzcd}
A \arrow[r, two heads, "\operatorname{coim}\phi"'] \arrow[rrr, bend left, "\phi"] & C \arrow[r, two heads, "\operatorname{coim}\underline{\phi}"] \arrow[rr, bend right, "\underline{\phi}"'] & C' \arrow[r] & B.
\end{tikzcd}$$

Since $\operatorname{coim}\phi$ is final with respect to factorizations such that the leading morphism is epi, there exists a unique morphism making the diagram

$$\begin{tikzcd}
 & C' \arrow[d, dashed, "\sigma"] \arrow[ddr, bend left, "\underline{\underline{\phi}}"] & \\
 & C \arrow[dr, "\underline{\phi}"] & \\
A \arrow[uur, two heads, bend left, "\operatorname{coim}\underline{\phi}\circ\operatorname{coim}\phi"] \arrow[ur, two heads, "\operatorname{coim}\phi"'] \arrow[rr, "\phi"'] & & B
\end{tikzcd}$$

commute. It follows by commutativity of the diagram that

$$\sigma \circ \operatorname{coim}\underline{\phi} \circ \operatorname{coim}\phi = 1_C \circ \operatorname{coim}\phi.$$

We insert the 1_C to make the following statement. Since $\operatorname{coim}\phi$ is epi, by definition 1.3 we must have $\sigma \circ \operatorname{coim}\underline{\phi} = 1_C$. Therefore, $\operatorname{coim}\underline{\phi}$ has a left-inverse, and is thus mono. Since A is abelian, $\operatorname{coim}\underline{\phi}$ is an isomorphism by lemma 1.5.

Note that since every kernel is the kernel of its cokernel,

$$\ker \operatorname{coker} \ker \phi = \ker \phi \implies \ker \operatorname{coim}\underline{\phi} = \ker \underline{\phi}.$$

Since $\underline{\phi}$ is an isomorphism,

$$0 = \ker \operatorname{coim}\underline{\phi} = \ker \underline{\phi}.$$

By lemma 1.3, $\underline{\phi}$ is a monomorphism. The statement for epimorphisms follows since A^{op} is abelian. ■

Our summary of the situation now becomes

$$\begin{tikzcd}
& K \arrow[dr, tail, bend left, "\operatorname{im}\phi"] & \\
A \arrow[ur, two heads, bend left, "\overline{\phi}"] \arrow[rr, "\phi"] \arrow[dr, two heads, bend right, "\operatorname{coim}\phi"'] & & B. \\
& C \arrow[ur, tail, bend right, "\underline{\phi}"'] &
\end{tikzcd}$$

Since the image and coimage are initial with respect to factorizations of this sort, the statement of our canonical decomposition becomesd the following.

Theorem 1.1 (Canonical Decomopsition)

Let $\phi : A \to B$ be a morphism in an abelian category. Then there exists a unique $\widetilde{\phi}$ such that

$$\begin{tikzcd}
A \arrow[r, two heads, "\operatorname{coim}\phi"'] \arrow[rrr, bend left, "\phi"] & C \arrow[r, tail, two heads, dashed, "\widetilde{\phi}"'] & K \arrow[r, tail, "\operatorname{im}\phi"'] & B
\end{tikzcd}$$

commutes.

Proof. It suffices to show the target of the cokernel is naturally isomorphic to the source of the kernel. By the universal properties of image and coimage, there exists a unique ψ making the diagram

$$\begin{tikzcd}
& C \arrow[ddr, tail, bend left, "\underline{\phi}"] & \\
& K \arrow[u, dashed, "\psi"] \arrow[dr, tail, "\operatorname{im}\phi"] & \\
A \arrow[uur, two heads, bend left, "\operatorname{coim}\phi"] \arrow[ur, two heads, "\overline{\phi}"] \arrow[rr, "\phi"] & & B
\end{tikzcd}$$

commute. By exercise 1, if $\psi \circ \phi$ is epi, then ψ is epi, and conversely if $\psi \circ \phi$ is mono, then ϕ is mono. Using this fact, note that $\psi \circ \overline{\phi}$ is epi, and thus ψ is epi. Similarly, note that $\underline{\phi} \circ \psi$ is mono by commutativity, and thus so is ψ. By lemma 1.5, ψ is an isomorphism, and by definition admits a unique inverse. Letting $\widetilde{\psi}$ be the inverse of ψ completes the proof. ■

1.2 Working in Abelian Categories

1.2.1 Exactness in Abelian Categories

We can now define what it means for a sequence to be exact in an abelian category. A sequence

$$0 \longrightarrow A \xrightarrow{\phi} B \xrightarrow{\psi} C \longrightarrow 0$$

is exact at B if $\psi \circ \phi = 0$, and $\operatorname{coker} \phi \circ \ker \psi = 0$. This first quality makes the sequence into a complex, and also tells us that ϕ factors through $\ker \psi$. Since $\ker \psi$ is a monomorphism, this yields a unique factorization of $\operatorname{im} \phi$ through $\ker \psi$. The second condition tells us that $\ker \psi$ factors through $\ker \operatorname{coker} \psi = \operatorname{im} \psi$, and thus we have $\operatorname{im} \phi = \ker \psi$. This recovers our definition of exactness in R-Mod, and also recovers some basic statements such as

- ψ is mono if and only if $\phi = 0$;
- ϕ is epi if and only if $\psi = 0$;
- The sequence is exact if and only if $\psi = \ker \phi$ and $\psi = \operatorname{coker} \phi$;
- The sequence $0 \to A \to B \to 0$ is exact if and only if $A \to B$ is an isomorphism.

For example, the natural sequence

$$0 \longrightarrow A \longrightarrow A \oplus B \longrightarrow B \longrightarrow 0$$

is exact.

1.2.2 The Snake Lemma, Again

We can now prove the snake lemma, purely arrow-theoretically. First, we need the following result.

Lemma 1.8

Let

$$\begin{tikzcd} A \times_C B \arrow[r, "\overline{\phi}"] \arrow[d, "\overline{\psi}"'] & B \arrow[d, "\psi"] \\ A \arrow[r, "\phi"'] & C \end{tikzcd}$$

be a pullback in an abelian category, and let ϕ be epi. Then $\overline{\phi}$ is epi. Similarly, if

$$\begin{tikzcd} C \arrow[r, "\psi"] \arrow[d, "\phi"'] & B \arrow[d, "\overline{\phi}"] \\ A \arrow[r, "\overline{\psi}"'] & A \amalg_C B \end{tikzcd}$$

is a pushout, and ψ is mono, then $\overline{\psi}$ is mono.

Proof. a ■

Proposition 1.2 (Snake Lemma)

Let

$$\begin{tikzcd}
0 \arrow[r] & L_1 \arrow[r, "\alpha_1"] \arrow[d, "\lambda"] & M_1 \arrow[r, "\beta_1"] \arrow[d, "\mu"] & N_1 \arrow[r] \arrow[d, "\nu"] & 0 \\
0 \arrow[r] & L_0 \arrow[r, "\alpha_0"] & M_0 \arrow[r, "\beta_0"] & N_0 \arrow[r] & 0
\end{tikzcd}$$

be a commutative diagram in an abelian category with exact rows. Then there exists an exact complex

$$0 \longrightarrow \mathrm{Ker}_\lambda \longrightarrow \mathrm{Ker}_\mu \longrightarrow \mathrm{Ker}_\nu \longrightarrow \mathrm{Cok}_\lambda \longrightarrow \mathrm{Cok}_\mu \longrightarrow \mathrm{Cok}_\nu \longrightarrow 0.$$

Proof. By assumption,

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & \mathrm{Ker}_\lambda \arrow[d, tail, "\ker\lambda"] & \mathrm{Ker}_\mu \arrow[d, tail, "\ker\mu"] & \mathrm{Ker}_\nu \arrow[d, tail, "\ker\nu"] & \\
0 \arrow[r] & L_1 \arrow[r, tail, "\alpha_1"] \arrow[d, "\lambda"] & M_1 \arrow[r, two heads, "\beta_1"] \arrow[d, "\mu"] & N_1 \arrow[r] \arrow[d, "\nu"] & 0 \\
0 \arrow[r] & L_0 \arrow[r, tail, "\alpha_0"] \arrow[d, two heads, "\operatorname{coker}\lambda"] & M_0 \arrow[r, two heads, "\beta_0"] \arrow[d, two heads, "\operatorname{coker}\mu"] & N_0 \arrow[r] \arrow[d, two heads, "\operatorname{coker}\nu"] & 0 \\
 & \mathrm{Cok}_\lambda \arrow[d] & \mathrm{Cok}_\mu \arrow[d] & \mathrm{Cok}_\nu \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

commutes. Consider the subdiagram

$$\begin{tikzcd}
\mathrm{Ker}_\lambda \arrow[d, "\ker\lambda"] & \mathrm{Ker}_\mu \arrow[d, "\ker\mu"] \\
L_1 \arrow[r, "\alpha_1"] \arrow[d, "\lambda"] & M_1 \arrow[d, "\mu"] \\
L_0 \arrow[r, "\alpha_0"] & M_0
\end{tikzcd}$$

Since $\lambda \circ \ker\lambda = 0$, it follows that $\alpha_0 \circ \lambda \circ \ker\lambda = 0$, and by commutivity we have $\mu \circ \alpha_1 \circ \ker\lambda = 0$. It follows by the universal property of kernels that $\alpha_1 \circ \ker\lambda$ factors uniquely through $\ker\mu$ such that the diagram

$$
\begin{tikzcd}
\mathrm{Ker}_\lambda \arrow[r, "\alpha_1 \circ \ker\lambda"] \arrow[rr, bend left, "0"] \arrow[dr, dashed] & M_1 \arrow[r, "\mu"] & M_0 \\
 & \mathrm{Ker}_\mu \arrow[u, "\ker\mu"] &
\end{tikzcd}
$$

commutes. Equivalent logic can be applied to find a unique morphism $\mathrm{Ker}_\mu \to \mathrm{Ker}_\nu$. Now consider the subdiagram

$$
\begin{tikzcd}
L_1 \arrow[r, "\alpha_1"] \arrow[d, "\lambda"] & M_1 \arrow[d, "\mu"] \\
L_0 \arrow[r, "\alpha_0"] \arrow[d, "\mathrm{coker}\,\lambda"] & M_0 \arrow[d, "\mathrm{coker}\,\mu"] \\
\mathrm{Cok}_\lambda & \mathrm{Cok}_\mu
\end{tikzcd}
$$

Since $\mathrm{coker}\,\mu \circ \mu = 0$, it follows that $\mathrm{coker}\,\mu \circ \mu \circ \alpha_1 = 0$, and by commutivity of the diagram we have $\mathrm{coker}\,\mu \circ \alpha_0 \circ \lambda = 0$. It follows by the universal property of cokernels that $\mathrm{coker}\,\mu \circ \alpha_0$ factors uniquely through $\mathrm{coker}\,\lambda$ such that the diagram

$$
\begin{tikzcd}
L_1 \arrow[r, "\lambda"] \arrow[rr, bend left, "0"] & L_0 \arrow[r, "\mathrm{coker}\,\mu\circ\alpha_0"] \arrow[d, "\mathrm{coker}\,\lambda"] & \mathrm{Cok}_\mu \\
 & \mathrm{Cok}_\lambda \arrow[ur, dashed] &
\end{tikzcd}
$$

commutes. By similar logic, we obtain a unique morphism $\mathrm{Cok}_\mu \to \mathrm{Cok}_\nu$ as well. Therefore, the diagram

$$
\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & \mathrm{Ker}_\lambda \arrow[r, dashed] \arrow[d, tail, "\ker\lambda"] & \mathrm{Ker}_\mu \arrow[r, dashed] \arrow[d, tail, "\ker\mu"] & \mathrm{Ker}_\nu \arrow[d, tail, "\ker\nu"] & \\
0 \arrow[r] & L_1 \arrow[r, tail, "\alpha_1"] \arrow[d, "\lambda"] & M_1 \arrow[r, two heads, "\beta_1"] \arrow[d, "\mu"] & N_1 \arrow[r] \arrow[d, "\nu"] & 0 \\
0 \arrow[r] & L_0 \arrow[r, tail, "\alpha_0"] \arrow[d, two heads, "\mathrm{coker}\,\lambda"] & M_0 \arrow[r, two heads, "\beta_0"] \arrow[d, two heads, "\mathrm{coker}\,\mu"] & N_0 \arrow[r] \arrow[d, two heads, "\mathrm{coker}\,\nu"] & 0 \\
 & \mathrm{Cok}_\lambda \arrow[r, dashed] \arrow[d] & \mathrm{Cok}_\mu \arrow[r, dashed] \arrow[d] & \mathrm{Cok}_\nu \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}
$$

commutes. We will show rows are exact later; first, we show a morphism $\mathrm{Ker}_\nu \to \mathrm{Cok}_\lambda$ exists.

Let

$$\begin{tikzcd}
M_1 \times_{N_1} \mathrm{Ker}_\nu \arrow[r, two heads, "\overline{\beta}_1"] \arrow[d] & \mathrm{Ker}_\nu \arrow[d, tail, "\ker\nu"] \\
M_1 \arrow[r, two heads, "\beta_1"] & N_1
\end{tikzcd}$$

be a pullback, where $\overline{\beta}_1$ is epi by lemma 1.8, and similarly let

$$\begin{tikzcd}
L_0 \arrow[r, tail, "\alpha_0"] \arrow[d, two heads, "\operatorname{coker}\lambda"] & M_0 \arrow[d] \\
\mathrm{Cok}_\lambda \arrow[r, tail, "\overline{\alpha}_0"] & \mathrm{Cok}_\lambda \amalg_{L_0} M_0
\end{tikzcd}$$

be a pushforward, where $\overline{\alpha}_0$ is mono by lemma 1.8. Note that

$$\begin{tikzcd}
L_1 \arrow[r, "0"] \arrow[d, tail, "\alpha_1"] \arrow[dr, "0"] & \mathrm{Ker}_\nu \arrow[d, tail, "\ker\nu"] \\
M_1 \arrow[r, two heads, "\beta_1"] & N_1
\end{tikzcd}$$

commutes by exactness of rows in the first diagram. Since the pullback is final with respect to this property, there exists a unique morphism σ making the diagram

$$\begin{tikzcd}
L_1 \arrow[dr, dashed, "\sigma"] \arrow[drr, bend left, "0"] \arrow[ddr, tail, bend right, "\alpha_1"'] & & \\
& M_1 \times_{N_1} \mathrm{Ker}_\nu \arrow[r, two heads, "\overline{\beta}_1"] \arrow[d] & \mathrm{Ker}_\nu \arrow[d, tail, "\ker\nu"] \\
& M_1 \arrow[r, two heads, "\beta_1"] & N_1
\end{tikzcd}$$

commute. Similarly, the diagram

$$\begin{tikzcd}
L_0 \arrow[r, two heads, "\operatorname{coker}\lambda"] \arrow[d, tail, "\alpha_0"] \arrow[dr, "0"] & \mathrm{Cok}_\lambda \arrow[d, "0"] \\
M_0 \arrow[r, two heads, "\beta_0"] & N_0
\end{tikzcd}$$

commutes by exactness of the first diagram, and since pushforwards are initial with respect to this property, there exists a unique morphism τ making the diagram

$$\begin{tikzcd}
L_0 \arrow[r, tail, "\alpha_0"] \arrow[d, two heads, "\operatorname{coker}\lambda"] & M_0 \arrow[d] \arrow[ddr, two heads, bend left, "\beta_0"] & \\
\mathrm{Cok}_\lambda \arrow[r, tail, "\overline{\alpha}_0"] \arrow[drr, bend right, "0"] & \mathrm{Cok}_\lambda \amalg_{L_0} M_0 \arrow[dr, dashed, "\tau"] & \\
 & & N_0
\end{tikzcd}$$

commute. It follows that the diagram

$$\begin{tikzcd}
0 \arrow[r] & L_1 \arrow[r, "\sigma"] \arrow[d, equal] & M_1 \times_{N_1} \mathrm{Ker}_\nu \arrow[r, two heads, "\overline{\beta}_1"] \arrow[d] \arrow[ddd, bend left, "\varepsilon"] & \mathrm{Ker}_\nu \arrow[r] \arrow[d, tail, "\ker\nu"] & 0 \\
0 \arrow[r] & L_1 \arrow[r, tail, "\alpha_1"] \arrow[d, "\lambda"] & M_1 \arrow[r, two heads, "\beta_1"] \arrow[d, "\mu"] & N_1 \arrow[r] \arrow[d, "\nu"] & 0 \\
0 \arrow[r] & L_0 \arrow[r, tail, "\alpha_0"] \arrow[d, two heads, "\operatorname{coker}\lambda"] & M_0 \arrow[r, two heads, "\beta_0"] \arrow[d] & N_0 \arrow[r] \arrow[d, equal] & 0 \\
0 \arrow[r] & \mathrm{Cok}_\lambda \arrow[r, tail, "\overline{\alpha}_0"] & \mathrm{Cok}_\lambda \amalg_{L_0} M_0 \arrow[r, "\tau"] & N_0 \arrow[r] & 0
\end{tikzcd}$$

commute. Moreover, we will be able to show that rows are exact. Note that by commutivity of the diagram, since $\operatorname{coker}\lambda \circ \lambda = 0$ and $\nu \circ \ker\nu = 0$, it follows that $\varepsilon \circ \sigma = 0$ and $\tau \circ \varepsilon = 0$.

WTS: $\operatorname{coker}\sigma = \ker\nu$, since

$$\begin{tikzcd}
 & \mathrm{Cok}_\sigma \arrow[drr, dashed] & & \mathrm{Cok}_\varepsilon \arrow[dr, dashed] & \\
L_1 \arrow[r, "\sigma"] \arrow[dr, dashed] & M_1 \times_{N_1} \mathrm{Ker}_\nu \arrow[r, "\varepsilon"] \arrow[u, "\operatorname{coker}\sigma"] \arrow[drr, dashed] & \mathrm{Cok}_\lambda \amalg_{L_0} M_0 \arrow[r, "\tau"] \arrow[ur, "\operatorname{coker}\varepsilon"] & & N_0 \\
 & \mathrm{Ker}_\varepsilon \arrow[u, "\ker\varepsilon"] & & \mathrm{Ker}_\tau \arrow[ul, "\ker\tau"] &
\end{tikzcd}$$

commutes. Wait this implies $\ker\nu$ is iso, and that happens iff $\nu = 0$ i think, so this no worky. ■

1.2.3 Working with 'Elements' in a Small Abelian Category

will do later

1.3 Complexes and Homology, Again

By the Freyd-Mitchell theorem, little will be lost by interpeting the following statements as occuring in a category such as R-Mod. I will do my best to keep the considerations as abstract as possible, however.

1.3.1 Reminder of Basic Definitions, General Strategy

A chain complex $(M_\bullet, d_\bullet)$, or simply $M_\bullet$, is a sequence in an abelian category

$$\dots \xrightarrow{d_{i+2}} M_{i+1} \xrightarrow{d_{i+1}} M_i \xrightarrow{d_i} M_{i-1} \xrightarrow{d_{i-1}} \dots$$

such that $d_{i+1} \circ d_i = 0$ for all i. A cochain complex $M^\bullet$ is a sequence

$$\dots \xrightarrow{d^{i-2}} M^{i-1} \xrightarrow{d^{i-1}} M^i \xrightarrow{d^i} M^{i+1} \xrightarrow{d^{i+1}} \dots$$

is a chain complex with ascending indicies. There is no change in the mathematics, so our choice of which to use is largely asthetic. The morphisms d are known as the *differentials*, and we will often denote by $d^\bullet_{M^\bullet}$, or simply by $d^\bullet$, the collection of all morphisms in the complex $M^\bullet$. In this chapter, we will work with cochain complexes and cohomology.

In categories with elements, we defined the homology

$$H^i(M^\bullet) \coloneqq \frac{\ker d^i}{\operatorname{im} d^{i-1}}.$$

However, this is not possible in the most general sense. To reconsile this definition, note that $\operatorname{im} d^{i-1} : I \to M^i$ factors uniquely through $\ker d^i$, giving a monomorphism $I \to \mathrm{Ker}_{d^i}$. Clearly, in a category with elements, the cokernel of this morphism gives us the cohomology, so we define it as such.

Definition 1.6: (Co)Homology

Let $M^\bullet$ be a chain complex in an abelian category. Then the cohomology $H^i(M^\bullet)$ of $M^\bullet$ at i is the target of the cokernel of the unique morphism ϕ making the diagram

$$\begin{tikzcd} I \arrow[r, "\operatorname{im} d^{i-1}"] \arrow[rr, bend left, "0"] \arrow[dr, dashed, "\phi"'] & M^i \arrow[r, "d^i"] & M^{i+1} \\ & \mathrm{Ker}_{d^i} \arrow[u, "\ker d^i"'] & \end{tikzcd}$$

commute.

I have sped forward slightly to also give a precise notion of resolution. A morphism $\alpha^\bullet$ in the category $\mathsf{C}(\mathsf{A})$ of chain complexes in A is a *quasi-isomorphism* if it induces an isomorphism in the cohomologies.

Definition 1.7: Resolution

A resolution of an object A is a complex $M^\bullet$ which is quasi-isomorphic to $\iota(A)$, which is defined as the complex

$$\dots \longrightarrow 0 \xrightarrow{d^1} A \xrightarrow{d^0} 0 \longrightarrow \dots$$

This means all cohomologies are zero, except for $H^0(M^\bullet)$, which is (viewed as an R-module) isomorphic to $\operatorname{coker} d^{-1}$, or is (viewed categorically) the target of $\operatorname{coker} d^{-1}$.

Resolutions become more useful if the M^is become special, eg. flat, projective, free, injective, etc. These allow us to associate to mathematical objects certian cochain complexes, which we can then take the cohomology of. We have seen this in the case of Tor and Ext, and it has much broader applications. One of our main goals here will be to show the resultant cohomologies of this construction are generally invariant under choice of the type of "special" objects to use in the resolution.

1.3.2 The Category of Complexes

Definition 1.8: Category of Complexes

The category $\mathsf{C}(\mathsf{A})$ of (co)chain complexes in A is defined as the category where

- $\mathrm{Obj}(\mathsf{C}(\mathsf{A}))$ consists of all cochain complexes in A;
- Morphisms $\alpha^\bullet : M^\bullet \to N^\bullet$ are collections of morphisms $\alpha^i : M^i \to N^i$ in A such that the diagram

$$\begin{tikzcd}
\dots \arrow[r, "d^{i-2}"] & M^{i-1} \arrow[r, "d^{i-1}"] \arrow[d, "\alpha^{i-1}"] & M^i \arrow[r, "d^i"] \arrow[d, "\alpha^i"] & M^{i+1} \arrow[r, "d^{i+1}"] \arrow[d, "\alpha^{i+1}"] & \dots \\
\dots \arrow[r, "d^{i-2}"] & N^{i-1} \arrow[r, "d^{i-1}"] & N^i \arrow[r, "d^i"] & N^{i+1} \arrow[r, "d^{i+1}"] & \dots
\end{tikzcd}$$

  commutes. That is,

$$\forall i : \alpha^{i+1} d^i_{M^\bullet} = d^i_{N^\bullet} \alpha^i.$$

The fact that this is a category is immediate by the fact that A is a category, and the fact that the diagram commutes. However, we can also say the following.

Lemma 1.9

$\mathsf{C}(\mathsf{A})$ is an abelian category.

The proof is not difficult but is long, and will be done in the exercises. This is the natural place to study resolutions of objects, as the category itself is abelian and thus we can talk about exact sequences of complexes, canonical decompositions, etc. Complexes of complexes will be important later. The important things to note is that kernels and cokernels are suitably defined as the collections of kernels; that is, $\ker \alpha^\bullet$ is simply the collection of maps $\ker \alpha^i$, and similarly for cokernels.

We will also use the following notation for various other categories:

- $\mathsf{C}^+(\mathsf{A})$ for the full subcategory of complexes bounded below (there exists n such that $L^i = 0$ for all $i \leq n$);
- $\mathsf{C}^-(\mathsf{A})$ for bounded above;
- $\mathsf{C}^{\geq 0}$ for complexes with $L^i = 0$ for all $i < 0$;
- $\mathsf{C}^{\leq 0}$;
- and we will also use P and I for projective/injective objects in A.

This new category gives us access to a few functors, such as $\iota_r : \mathsf{A} \to \mathsf{C}(\mathsf{A})$ which is defined for any $r \in \mathbb{Z}$, which maps A to the complex of 0s, except for a single A at the rth spot. We generally identify A inside of $\mathsf{C}(\mathsf{A})$ with it's image $\iota_0(\mathsf{A})$, which we simply denote ι.

We also have shift functors $\mathsf{C}(\mathsf{A}) \to \mathsf{C}(\mathsf{A})$ defined by $M^\bullet \mapsto M[r]^\bullet$, where $M[r]^i := M^{i+r}$ and $d^i_{M[r]^\bullet} := (-1)^r d^{i+r}_{M^\bullet}$. Additionally. cohomology is a functor, and in fact an important one.

Lemma 1.10

Cohomology $M^\bullet \mapsto H^i(M^\bullet)$ is an additive covariant functor $\mathsf{C}(\mathsf{A}) \to \mathsf{A}$.

Proof. This means that the functorial relation

$$\mathrm{Hom}_{\mathsf{C(A)}}(M^\bullet, N^\bullet) \to \mathrm{Hom}_{\mathsf{A}}(H^i(M^\bullet), H^i(N^\bullet))$$

is also a homomorphism of abelian groups. The action on morphisms can be defined as follows. Let

$$\begin{tikzcd}
M^{i-1} \arrow[r, "d^{i-1}"] \arrow[d, "\alpha^{i-1}"] & M^i \arrow[r, "d^i"] \arrow[d, "\alpha^i"] & M^{i+1} \arrow[d, "\alpha^{i-1}"] \\
N^{i-1} \arrow[r, "d^{i-1}"] & N^i \arrow[r, "d^i"] & N^{i+1}
\end{tikzcd}$$

be a commutative subdiagram of the diagram induced by the morphism $\alpha^\bullet : M^\bullet \to N^\bullet$. Recall our definition of the homology as the cokernels:

$$\begin{tikzcd}
\mathrm{Im}\, d_M^{i-1} \arrow[r, dashed, "\phi_M"] \arrow[dr, tail, "\mathrm{im}\, d_M^{i-1}"] & \mathrm{Ker}\, d_M^i \arrow[r, two heads, "\mathrm{coker}\, \phi_M"] \arrow[d, tail, "\ker d_M^i"] & H^i(M^\bullet) \\
M^{i-1} \arrow[u, dashed, "\psi_M"] \arrow[r, "d_M^{i-1}"] \arrow[d, "\alpha^{i-1}"] & M^i \arrow[r, "d_M^i"] \arrow[d, "\alpha^i"] & M^{i+1} \arrow[d, "\alpha^{i-1}"] \\
N^{i-1} \arrow[r, "d_N^{i-1}"] \arrow[d, dashed, "\psi_B"] & N^i \arrow[r, "d_N^i"] & N^{i+1} \\
\mathrm{Im}\, d_N^{i-1} \arrow[r, dashed, "\phi_N"'] \arrow[ur, tail, "\mathrm{im}\, d_N^{i-1}"] & \mathrm{Ker}\, d_N^i \arrow[r, two heads, "\mathrm{coker}\, \phi_N"'] \arrow[u, tail, "\ker d_N^i"] & H^i(N^\bullet)
\end{tikzcd}$$

By the universal properties of kernels, cokernels, and images, this diagram commutes. By commutivity of the diagram,

$$d_N^i \circ \alpha^i \circ \ker d_M^i = \alpha^{i-1} \circ d_M^i \circ \ker d_M^i = 0,$$

and thus the composition $\alpha^i \circ \ker d_M^i$ factors uniquely through $\ker d_N^i$ such that the diagram

$$\begin{tikzcd}
\mathrm{Ker}\, d_M^i \arrow[r, "\alpha^i \ker d_M^i"] \arrow[rr, bend left, "0"] \arrow[dr, dashed, "\sigma"'] & N^i \arrow[r, "d_N^i"] & N^{i+1} \\
& \mathrm{Ker}\, d_N^i \arrow[u, tail, "\ker d_N^i"] &
\end{tikzcd}$$

commutes. We thus have an induced map $\sigma : \mathrm{Ker}\, d_M^i \to \mathrm{Ker}\, d_N^i$.

By lemma 1.7, ψ_A and ψ_B are epi, since they are induced by the image. Note that with the addition of σ, by commutivity of the diagram we have

$$\begin{tikzcd}
M^{i-1} \arrow[r, dashed, two heads, "\psi_M"] \arrow[d, "\alpha^{i-1}"] & \operatorname{Im} d_M^{i-1} \arrow[r, dashed, "\phi_A"] & \operatorname{Ker} d_M^i \arrow[r, tail, "\ker d_M^i"] \arrow[d, dashed, "\sigma"] & A_i \arrow[r, "\alpha^i"] & B_i \\
N^{i-1} \arrow[r, dashed, two heads, "\psi_N"] & \operatorname{Im} d_N^{i-1} \arrow[r, dashed, "\phi_N"] \arrow[rr, bend right, "0"'] & \operatorname{Ker} d_N^i \arrow[r, two heads, "\operatorname{coker} \phi_N"'] \arrow[urr, tail, "\ker d_N^i"'] & H^i(B^\bullet) &
\end{tikzcd}$$

It follows that

$$\operatorname{coker} \phi_N \circ \phi_N \circ \psi_N \circ \alpha^{i-1} = 0.$$

By commutivity of the diagram,

$$\operatorname{coker} \phi_N \circ \sigma \circ \phi_M \circ \psi_M = 0.$$

Since ψ_M is epi, by lemma 1.1, it follows that

$$\operatorname{coker} \phi_N \circ \sigma \circ \phi_M = 0.$$

Therefore, $\operatorname{coker} \phi_N \circ \sigma$ factors uniquely through $\operatorname{coker} \phi_M$:

$$\begin{tikzcd}
\operatorname{Im} d_M^{i-1} \arrow[r, "\phi_M"] \arrow[rrr, bend left, "0"] & \operatorname{Ker} d_M^i \arrow[r, "\sigma"] \arrow[d, two heads, "\operatorname{coker} \phi_A"'] & \operatorname{Ker} d_N^i \arrow[r, "\operatorname{coker} \phi_N"] & H^i(N^\bullet) \\
& H^i(A^\bullet) \arrow[urr, dashed, "H^i(\alpha^\bullet)"'] & &
\end{tikzcd}$$

commutes by the universal property of cokernels. Therefore, $H^i(-)$ uniquely induces a morphism $H^i(\alpha^\bullet) : H^i(M^\bullet) \to H^i(N^\bullet)$.

It is easily seen that this assignment is functorial by considering commutivity of the diagram induced by a chain of morphisms $L^\bullet \to M^\bullet \to N^\bullet$, and noting that $H^i(\beta^\bullet) \circ H^i(\alpha^\bullet)$ corresponds to doing this process to each subdiagram, which can be attached to each other by commutivity to create the equivalent morphism $H^i(\beta^\bullet \alpha^\bullet)$. Since this assignment is natural, apparently that means it's also an abelian group homomorphism. ■

We can also view cohomology as a functor $\mathsf{C}(\mathsf{A}) \to \mathsf{C}(\mathsf{A})$, by taking the cohomology $H^i(M^\bullet)$ and placing it at degree i. Since the resulting cohomology of this sequence is 0 except at i, where it is $H^i(M^\bullet)$ once again, we can form a new complex $H^\bullet(M^\bullet)$ with

$$H^\bullet(H^\bullet(M^\bullet)) = H^\bullet(M^\bullet).$$

1.3.3 The Long Exact Cohomology Sequence

This allows us to reconsider the snake lemma *again*. Written cohomologically, the snake lemma is the commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & L^0 \arrow[r, "\alpha^0"] & M^0 \arrow[r, "\beta^0"] & N^0 \arrow[r] & 0 \\
0 \arrow[r] & L^1 \arrow[r, "\alpha^1"] \arrow[u, "\lambda^0"] & M^1 \arrow[r, "\beta^1"] \arrow[u, "\mu^0"] & N^1 \arrow[r] \arrow[u, "\nu^0"] & 0
\end{tikzcd}$$

with exact rows. We can view each column as a complex

$$L^\bullet : \qquad \dots \longrightarrow 0 \longrightarrow L^0 \xrightarrow{\lambda^0} L^1 \longrightarrow 0 \longrightarrow \dots,$$

and then the snake lemma is nothing more than the exact sequence

$$0 \longrightarrow L^\bullet \longrightarrow M^\bullet \longrightarrow N^\bullet \longrightarrow 0$$

in $\mathsf{C}(\mathsf{A})$. The snake lemma then gives us an exact sequence

$$\begin{tikzcd}
0 \arrow[r] & H^0(L^\bullet) \arrow[r] & H^0(M^\bullet) \arrow[r] & H^0(N^\bullet) \arrow[dll] & \\
& H^1(L^\bullet) \arrow[r] & H^1(M^\bullet) \arrow[r] & H^1(N^\bullet) \arrow[r] & 0
\end{tikzcd}$$

This works since, for example, the image $0 \to L^0$ is simply the 0 morphism, and factoring it through $\ker \lambda$ we get the diagram

$$\begin{tikzcd}
\operatorname{Im} 0 \arrow[r, dashed, "\phi"] \arrow[dr, "\operatorname{im} 0"] & \operatorname{Ker} \lambda \arrow[r, "\operatorname{coker} \phi"] \arrow[d, "\ker \lambda"] & \operatorname{Ker} \lambda / 0 \\
0 \arrow[u, dashed] \arrow[r, "0"] & L^0 &
\end{tikzcd}$$

It can be shown fairly easily that $\operatorname{coker} \phi$ is an isomorphism here. Similarly, we obtain the cokernel when taking $H^1(-)$, giving us the snake lemma.

This can be very naturally generalized to any short exact sequence

$$S^\bullet : \qquad 0 \longrightarrow L^\bullet \xrightarrow{\alpha^\bullet} M^\bullet \xrightarrow{\beta^\bullet} N^\bullet \longrightarrow 0$$

in $\mathsf{C}(\mathsf{A})$, where the sequences $L^\bullet, M^\bullet, N^\bullet$ are of arbitrary length. This can be viewed as a commutative diagram in A:

$$\begin{tikzcd}
& \vdots & \vdots & \vdots & \\
0 \arrow[r] & L^{i+1} \arrow[r] \arrow[u] & M^{i+1} \arrow[r] \arrow[u] & N^{i+1} \arrow[r] \arrow[u] & 0 \\
0 \arrow[r] & L^{i} \arrow[r] \arrow[u] & M^{i} \arrow[r] \arrow[u] & N^{i} \arrow[r] \arrow[u] & 0 \\
0 \arrow[r] & L^{i-1} \arrow[r] \arrow[u] & M^{i-1} \arrow[r] \arrow[u] & N^{i-1} \arrow[r] \arrow[u] & 0 \\
& \vdots \arrow[u] & \vdots \arrow[u] & \vdots \arrow[u] &
\end{tikzcd}$$

Rows are exact here, since $S^\bullet$ is exact. We can perform a similar construction as to the snaking homomorphism in the Snake lemma, which can be manipulated to induce a morphism $\delta^i : H^i(N^\bullet) \to H^{i+1}(L^\bullet)$ (ill show this later). Since each H^i is a functor, there are natural morphisms $H^i(L^\bullet) \to H^i(M^\bullet) \to H^i(N^\bullet)$ for all i, and thus induces a long complex

$$\dots \longrightarrow H^{i-1}(N^\bullet) \xrightarrow{\delta^{i-1}} H^i(L^\bullet) \longrightarrow H^i(M^\bullet) \longrightarrow H^i(N^\bullet) \longrightarrow \dots$$

The following theorem shows that this sequence generalizes the snake lemma.

Theorem 1.2

The above sequence is exact.

Proof. do later ■

1.3.4 Triangles

We can immediately connect $N^\bullet$ to $L^\bullet$ by the 0 morphism

$$N^\bullet \xrightarrow{0} L^\bullet,$$

and thus the exactness of the original short exact sequence is the same as the exactness of the sequences

$$N^\bullet \xrightarrow{0} L^\bullet \longrightarrow M^\bullet,$$

$$M^\bullet \longrightarrow N^\bullet \xrightarrow{0} L^\bullet,$$

$$L^\bullet \longrightarrow M^\bullet \longrightarrow N^\bullet$$

at their centers. We write this as an exact triangle

$$\begin{tikzcd} & L^\bullet \arrow[dr] & \\ N^\bullet \arrow[ur, "+1"] & & M^\bullet \arrow[ll] \end{tikzcd}$$

and we view $N^\bullet \to L^\bullet$ as a morphism $N^\bullet \to L^\bullet[1]$. Another common notation is

$$L^\bullet \longrightarrow M^\bullet \longrightarrow N^\bullet \xrightarrow{+1}$$

This is the same as the commutative diagram

$$\begin{tikzcd} & L^{i+1} \arrow[dr] & \\ N^{i+1} & & M^{i+1} \arrow[ll] \\ & L^i \arrow[uu] \arrow[dr] & \\ N^i \arrow[uu] \arrow[uuur, "0"] & & M^i \arrow[ll] \arrow[uu] \\ & L^{i-1} \arrow[uu] \arrow[dr] & \\ N^{i-1} \arrow[uu] \arrow[uuur, "0"] & & M^{i-1} \arrow[ll] \arrow[uu] \end{tikzcd}$$

where triangles are exact. Using theorem **??**, we can generalize this construction more naturally by taking the exact triangle

$$\begin{tikzcd}
& H^\bullet(L^\bullet) \arrow[dr] & \\
H^\bullet(N^\bullet) \arrow[ur, "+1"] & & H^\bullet(M^\bullet) \arrow[ll]
\end{tikzcd}$$

The stepping morphism is the connecting morphism explored earlier, and thus the long exact cohomology sequence actually sends the original exact triangle where the connecting morphism is 0, to the exact triangle of cohomologies:

$$\begin{tikzcd}
& & L^\bullet \arrow[dr] & & & & H^\bullet(L^\bullet) \arrow[dr] & \\
H^\bullet : & N^\bullet \arrow[ur, "+1"] & & M^\bullet \arrow[ll] & \arrow[r, mapsto] & H^\bullet(N^\bullet) \arrow[ur, "+1"] & & H^\bullet(M^\bullet) \arrow[ll]
\end{tikzcd}$$

There is clearly something important here, however it is unclear why applying the cohomology functor should induce the connecting morphism. We will see later that "distinguished" (sometimes called exact, but not exact in the context of this conversation) triangles lead to long exact cohomological sequences more directly, and these triangles satisfy a number of axioms. This will lead to the exploration of the category $\mathsf{K}(\mathsf{A})$, which will be a *triangulated category*. Further explorations of this topic, however, are beyond the scope of this book.

1.4 Cones and Homotopies

If $\alpha^\bullet$ is a morphism $M^\bullet \to N^\bullet$, it is natural to consider the induced morphism $H^\bullet(\alpha^\bullet) : H^\bullet(M^\bullet) \to H^\bullet(N^\bullet)$, where each α^i is the morphism $H^i(M^\bullet) \to H^i(N^\bullet)$ described in the proof of lemma **??**. However, $H^\bullet$ is in general not exact, and this makes it difficult to describe the kernel and cokernel of $H^\bullet(\alpha^\bullet)$ in terms of $\alpha^\bullet$. We will use the long exact cohomology sequence to account for this, and will also begin exploring "homotopies", which are so important that we will end up creating a brand new homotopic category of complexes.

1.4.1 The Mapping Cone of a Morphism

Given a morphism $\alpha^\bullet : M^\bullet \to N^\bullet$, the mapping cone $MC(\alpha)^\bullet$ will recover $H^\bullet(\alpha^\bullet)$ as the connecting morphism in the long cohomology sequence, will define the third vertex in a special triangle

$$\begin{tikzcd}
& M^\bullet \arrow[dr] & \\
L[1]^\bullet \arrow[ur, "0;+1"] & & MC(\alpha)^\bullet \arrow[ll]
\end{tikzcd}$$

such that the triangle induced by cohomology

$$\begin{tikzcd}
& H^\bullet(M^\bullet) \arrow[dr] & \\
H^\bullet(L[1]^\bullet) \arrow[ur, "+1"] & & H^\bullet(MC(\alpha)^\bullet) \arrow[ll]
\end{tikzcd}$$

will have the connecting morphisms given by $\alpha^\bullet : H^\bullet(L[1]^\bullet) \to H^\bullet(M^\bullet)$. That is, although morphisms induced by cohomology cannot be extended to exact sequences, they *can* be extended to exact triangles by including the mapping cone.

Definition 1.9: Mapping Cone

The mapping cone $MC(\alpha)^\bullet$ of a morphism $\alpha^\bullet : M^\bullet \to N^\bullet$ is defined as the complex consisting of objects

$$MC(\alpha)^i := L[1]^i \oplus M^i,$$

and morphisms $L[1]^i \oplus M^i \to L[1]^{i+1} \oplus M^{i+1}$ are given by the unique morphism making the diagram

$$\begin{tikzcd}
 & & L[1]^{i+1} \\
L[1]^i \oplus M^i \arrow[r, dashed, "d^i_{MC(\alpha)}"] \arrow[urr, bend left, "-d^i_{[L]}"] \arrow[drr, bend right, "(\alpha^{i+1} \circ \pi_{L[1]^i}) + d^i_M"'] & L[1]^{i+1} \oplus M^{i+1} \arrow[ur, two heads, "\pi^{i+1}_{L[1]}"] \arrow[dr, two heads, "\pi_{M^{i+1}}"] & \\
 & & M^{i+1}
\end{tikzcd}$$

commute by the universal property of products.

More conventionally, we would write

$$d^i_{MC(\alpha)} := \left(-d^i_{[L]}\right) \times \left(\left(\alpha^{i+1} \circ \pi_{L[1]^i}\right) + d^i_M\right).$$

By definition, we have split exact sequences

$$0 \longrightarrow L[1]^i \longrightarrow L[1]^i \oplus M^i \longrightarrow M^i \longrightarrow 0$$

for all i, and thus the sequence

$$0 \longrightarrow L[1]^\bullet \longrightarrow MC(\alpha)^\bullet \longrightarrow M^\bullet \longrightarrow 0$$

is exact. Now we can show the mapping cone satisfies the properties described above.

Proposition 1.3

There is an exact triangle

$$\begin{tikzcd}
 & H^\bullet(M^\bullet) \arrow[dr] & \\
H^\bullet(L[1]^\bullet) \arrow[ur, "+1"] & & H^\bullet(MC(\alpha)^\bullet) \arrow[ll]
\end{tikzcd}$$

where the connecting morphism is $H^\bullet(\alpha^\bullet)$.

Proof. The triangle exists by theorem 1.2, and by our discussion above, rows centered at the mapping cone are exact. It suffices to show that the induced morphism is α, since exactness follows in general from the theorem as well due to functorality of cohomology. By the logic of the snake lemma, it suffices to show the connecting morphism is α. This is easily shown by the same proof as in the snake lemma. ■

Corollary 1.1

Let $\alpha : M \to N$ be a morphism of cochain complexes. Then the induced morphism $H(M) \to H(L)$ is an isomorphism if and only if the mapping cone $MC(\alpha)$ is exact.

1.4.2 Quasi-Isomorphisms and Derived Categories

Definition 1.10: Quasi-Isomorphism

A quasi-isomorphism $\alpha^\bullet$ is a morphism $M^\bullet \to N^\bullet$ that induces an isomorphism $H^\bullet(M^\bullet) \to M^\bullet(N^\bullet)$ in cohomology.

By the previous corrollary, this condition is equivalent to stating that $MC(\alpha^\bullet)$ is exact. Recall that a resolution of an object A is a complex $M^\bullet$ that is quasi-isomorphic to $\iota(A)$; that is, all cohomologies are 0, except for $H^0(M^\bullet)$. This setup is given by the diagram

$$\begin{tikzcd} \dots \arrow[r] & M^{-1} \arrow[r, "d^{-1}"] \arrow[d] & M^0 \arrow[r, "d^0"] \arrow[d] & M^1 \arrow[r, "d^1"] \arrow[d] & \dots \\ \dots \arrow[r] & 0 \arrow[r] & A \arrow[r] & 0 \arrow[r] & \dots \end{tikzcd}$$

and the only nontrivial vertical map must be defined as $\operatorname{coker} d^{-1}$. Quasi-isomorphisms are in this sense generalizations of resolutions.

It would be nice if quasi-isomorphisms were invertible, since then the isomorphism classes of complexes would be encoded in their cohomology. This is in general not the case however, and they do not define an equivalence relation at all. However, identifying all chain complexes with an equivalence class and then quotienting is *not* the main goal of quotients; the main goal is studying the factorizations of maps through these quotients: functions with the same action on elements. It is thus not our goal to identify all complexes linked by a quasi-isomorphism, but rather to move information through them. The natural place to do this is in a category where they *are* invertible.

To do this, we can study additive functors $\mathscr{F} : \mathsf{C}(\mathsf{A}) \to \mathsf{D}$ such that $\mathscr{F}(\alpha^\bullet)$ is an isomorphism for all quasi-isomorphisms $\alpha^\bullet$. Clearly, cohomology is one such functor. It is most natural to phrase this as a universal problem, where we are looking for a category $\mathsf{D}(\mathsf{A})$ satisfying

$$\begin{tikzcd} \mathsf{C} \arrow[r] \arrow[d] & \mathsf{D} \\ \mathsf{D}(\mathsf{A}) \arrow[ur, dashed] & \end{tikzcd}$$

Such a construction exists if localization can be carried out, but such constructions are beyond this book, and are best suited for a more advanced text. For this book, we will simply take this category to exist and will ignore the specifics. We call this the *derived category*. We will be able to realize full definitions in some specific cases, however.

The derived category is not abelian, but enough structure remains to do homological algebra. Objects in $\mathsf{D}(\mathsf{A})$ have cohomology, and there exist "distinguished triangles", which abstract exact sequences and give long exact cohomology sequences. This makes it a *triangulated category*. And of course, it has applications in fucking string theory of all things.

To understand the abstract counterintuitive nature of $\mathsf{D}(\mathsf{A})$, note that if $M^\bullet$ is exact, then $H^\bullet(M^\bullet) = 0$, meaning the 0 morphism $0 \to M^\bullet$ is a quasi-isomorphism, and thus invertible in $\mathsf{D}(\mathsf{A})$. This means that nonzero complexes in $\mathsf{C}(\mathsf{A})$ may become 0 in $\mathsf{D}(\mathsf{A})$.

Lemma 1.11

Let $\mathscr{F} : \mathsf{C}(\mathsf{A}) \to \mathsf{D}$ be an additive functor that maps quasi-isomorphisms to isomorphisms.

- If $M^\bullet$ is exact, then $\mathscr{F}(M^\bullet) = 0$.
- If $\alpha^\bullet : L^\bullet \to M^\bullet$ admits a factorization

$$\begin{tikzcd} L^\bullet \arrow[rr, "\alpha^\bullet"] \arrow[dr] & & N^\bullet \\ & M^\bullet \arrow[ur] & \end{tikzcd}$$

in such a way that $M^\bullet$ is exact, then $\mathscr{F}(\alpha^\bullet) = 0$.

Proof. It is obvious that the second statement follows from the first, as we must have $\mathscr{F}(M^\bullet) = 0$ by the first, and thus

$$\begin{tikzcd} \mathscr{F}(L^\bullet) \arrow[rr, "\alpha^\bullet"] \arrow[dr] & & \mathscr{F}(N^\bullet) \\ & 0 \arrow[ur] & \end{tikzcd}$$

commutes.

For the first claim, note that since $\mathscr{F}$ is additive, $\mathscr{F}(0) = 0$. It follows that the diagram

$$\begin{tikzcd} \mathscr{F}(M^\bullet) \arrow[r, "\mathscr{F}(0)"] \arrow[rr, bend left, "1_{\mathscr{F}(M^\bullet)}"] & \mathscr{F}(M^\bullet) \arrow[r, "\mathscr{F}(0)^{-1}"] & \mathscr{F}(M^\bullet) \end{tikzcd}$$

commutes, and thus $1_{\mathscr{F}(M^\bullet)} = 0 \circ 0 = 0$. We have that $\mathrm{Hom}_{\mathsf{D}}(\mathscr{F}(M^\bullet), \mathscr{F}(M^\bullet))$ must be a singleton by a generalization of the $0 \neq 1$ in nontrivial rings proof to more general ring-like structures. ■

Constructions of derived categories and triangulated categories are once again beyond the scope of this book, so we will simply assume they exist.

1.4.3 Homotopy

Quasi-isomorphisms seem difficult to work with, so we again take inspiration from topology and introduce the notion of homotopy in an algebraic setting.

Definition 1.11: Homotopy

A homotopy between two morphisms $\alpha^\bullet, \beta^\bullet : L^\bullet \to M^\bullet$ is a collection of maps $h^i : L^i \to M^{i-1}$ such that for all i,

$$\beta^i - \alpha^i = d_M^{i-1} h^i + h^{i+1} d_L^i.$$

We say $\alpha^\bullet$ is homotopic to $\beta^\bullet$ and write $\alpha^\bullet \sim \beta^\bullet$.

Since in general $h^\bullet$ does not form a morphism $L^\bullet \to M[-1]^\bullet$, the diagram formed by this relation is in general non-commutative, meaning homotopy is difficult to visualize.

Definition 1.12: Homotopy Equivalence

A morphism $\alpha^\bullet : L^\bullet \to M^\bullet$ is a *homotopy equivalence* if there exists $\beta^\bullet : M^\bullet \to L^\bullet$ such that $\alpha^\bullet\beta^\bullet \sim 1_{M^\bullet}$ and $\beta^\bullet\alpha^\bullet \sim 1_{L^\bullet}$. The complexes $L^\bullet, M^\bullet$ are said to be homotopy equivalent if there exists a homotopy equivalence $M^\bullet \to L^\bullet$.

It is immediately checked that $\sim$ and homotopy equivalence are equivalence relations.

Proposition 1.4

If $\alpha^\bullet, \beta^\bullet : L^\bullet \to M^\bullet$ are homotopic morphisms of cochain complexes, then they induce the same morphism on cohomology; that is, $H^\bullet(\alpha^\bullet) = H^\bullet(\beta^\bullet)$.

Proof. Working through the arrow-theoretic version is hard and i may yield ∎

Corollary 1.2

Homotopy equivalent complexes have isomorphic cohomology.

Proof. Since $\alpha^\bullet \circ \beta^\bullet$ and $\beta^\bullet \circ \alpha^\bullet$ are homotopic to the identity, by proposition 1.4 they induce the same morphism as the identity in cohomology, and are thus inverses in cohomology. ∎

In general, homotopy equivalence is a stronger requirement than quasi-isomorphism, so this does not classify quasi-isomorphisms, although we clearly have that homotopy equivalent morphisms are quasi-isomorphisms.

Homotopy is extremely important, as homotopic complexes mean that the cohomology is independent of the choice of morphism, paving the way for functors such as the Ext and Tor functors.

Lemma 1.12

Let $\mathscr{F} : \mathsf{A} \to \mathsf{B}$ be an additive functor between abelian categories, and let $\mathsf{C}(\mathscr{F}) : \mathsf{C}(\mathsf{A}) \to \mathsf{C}(\mathsf{B})$ be the induced functor. If $\alpha^\bullet \sim \beta^\bullet$ in $\mathsf{C}(\mathsf{A})$, then $\mathsf{C}(\mathscr{F})(\alpha^\bullet) \sin \mathsf{C}(\mathscr{F})(\beta^\bullet)$ in $\mathsf{C}(\mathsf{D})$, and if $L^\bullet$ and $M^\bullet$ are homotopy equivalent, then so are $\mathsf{C}(\mathscr{F})(L^\bullet)$ and $\mathsf{C}(\mathscr{F})(M^\bullet)$.

Proof. The second statement follows from the first. Since $\mathscr{F}$ is not just additive on $\mathsf{C}(\mathsf{A})$ but also on A, we have $\mathscr{F}(\beta^i) - \mathscr{F}(\alpha^i) = \mathscr{F}(d_M^{i-1})\mathscr{F}(h^i) + \mathscr{F}(h^{i+1})\mathscr{F}(d_L^i)$. This satisfies the definition of a homotopy. ∎

Theorem 1.3

Let $\mathscr{F} : \mathsf{A} \to \mathsf{B}$ be an additive functor. If $L^\bullet$ and $M^\bullet$ are homotopy equivalent, then there exists an isomorphism

$$H^\bullet(\mathsf{C}(\mathscr{F})(L^\bullet)) \cong H^\bullet(\mathsf{C}(\mathscr{F})(M^\bullet)).$$

Proof. This is immediate after the previous results. By lemma 1.12, $\mathsf{C}(\mathscr{F})(L^\bullet)$ is homotopy equivalent to $\mathsf{C}(\mathscr{F})(M^\bullet)$. By corollary 1.2, they are thus isomorphic. ∎

The book allows this statement to be a theorem, since it shows that if a mechanism for producing cochain complexes can be defined *up to homotopy*, then cohomology induces an interesting invariant.

1.5 The Homotopic Category. Complexes of Projectives and Injectives

The moral of the previous section is that homotopy equivalent complexes have the same cohomology (as do any quasi-isomorphic complexes), and thus an important step towards undersanding the inversion of all quasi-isomorphisms will be the homotopic category, which inverts homotopy equivalences.

1.5.1 Homotopic Maps are Identified in the Derived Category

Do all additive functors $\mathscr{F} : \mathsf{C}(\mathsf{A}) \to \mathsf{D}$ that invert quasi-isomorphisms identify homotopic morphisms, like cohomology?

Lemma 1.13

Let $\mathscr{F} : \mathsf{C}(\mathsf{A}) \to \mathsf{D}$ be an additive functor that inverts quasi-isomorphisms. Then for any homotopic morphisms $\alpha^\bullet, \beta^\bullet : L^\bullet \to M^\bullet$, we have $\mathscr{F}(\alpha^\bullet) = \mathscr{F}(\beta^\bullet)$.

Proof. It suffices to show if $a^\bullet \sim 0$, then $\mathscr{F}(\alpha^\bullet) = 0$. By a previous lemma, it suffices further to show $a^\bullet$ factors through an exact complex; we will use $MC(1_{L^\bullet})^\bullet$. Recall our definition of the morphisms in $MC(1_{L^\bullet})^\bullet$ as

$$d^i_{MC(1_{L^\bullet})} = \left(-d^i_{L[1]}\right) \times \left(1^{i+1}_{L^\bullet}\pi_{L[1]} + d^i_M\right).$$

Since the identity is clearly a quasi-isomorphism, $MC(1_{L^\bullet})^\bullet$ is exact. By the universal property of coproducts, there exists an injection $i^\bullet : L^\bullet \to L[1]^\bullet \oplus L^\bullet = MC(1_{L^\bullet})^\bullet$. Before we can define a second map, recall that $\alpha^\bullet \sim 0$, there exist maps $h^\bullet : L \to M[-1]$ such that

$$\alpha^\bullet = d_M^{\bullet-1}h^\bullet + h^{\bullet+1}d_L^\bullet.$$

To define a second map $\pi^\bullet : MC(1_{L^\bullet})^\bullet \to M^\bullet$, let

$$\pi^\bullet = h^{\bullet+1}\pi_{L[1]} + \alpha^\bullet\pi_L^\bullet.$$

Viewed as a diagram, this looks like the sum of the morphisms

$$\begin{tikzcd}
& & L[1]^\bullet \arrow[r, "h^{\bullet+1}"] & M^\bullet \\
L^\bullet \arrow[r, dashed, "i^\bullet"] \arrow[urr, bend left, "0"] \arrow[drr, bend right, "1_{L^\bullet}"'] & L[1]^\bullet \oplus L^\bullet \arrow[ur, "\pi^\bullet_{L[1]}"] \arrow[dr, "\pi^\bullet_L"] & & \\
& & L \arrow[r, "\alpha^\bullet"] & M^\bullet
\end{tikzcd}$$

By the universal property of products, this diagram commutes, so it is clear that the composition $\pi^\bullet i^\bullet = \alpha^\bullet$. By lemma 1.11, since $\alpha^\bullet$ factors through an exact complex, $\mathscr{F}(\alpha^\bullet) = 0$. ■

This tells us that every additive functor that identifies homotopic morphisms must factor through the category obtained by identifying these morphisms.

1.5.2 Definition of the Homotopic Category of Complexes

Definition 1.13: Homotopic Category

Let A be abelian. The homotopic category $\mathsf{K}(\mathsf{A})$ of cochain complexes in A is the category where $\mathrm{Obj}(\mathsf{C}(\mathsf{A})) = \mathrm{Obj}(\mathsf{K}(\mathsf{A}))$, and

$$\mathrm{Hom}_{\mathsf{K}(\mathsf{A})}(A, B) = \mathrm{Hom}_{\mathsf{C}(\mathsf{A})}(A, B)/\sim,$$

where $\sim$ is the homotopy relation.

This is a category, since $\sim$ respects function composition as an equivalence relation. There also exist bounded version $\mathsf{K}^+(\mathsf{A})$, $\mathsf{K}^-(\mathsf{A})$, and other variants such as projective and injective variants. This category does not have abelian structure, since homotopy does not preserve kernels and cokernels, but we can at least say the following.

Lemma 1.14

$\mathsf{K}(\mathsf{A})$ is an additive category.

The motivation for the homotopic category is to invert homotopy equivalences. Indeed, there is a naturak identity functor $\mathsf{C}(\mathsf{A}) \to \mathsf{K}(\mathsf{A})$ mapping morphisms to their homotopy class. This allows lemma **??** to be reinterpreted as saying $\mathsf{K}(\mathsf{A})$ is universal in the following way: any additive functor $\mathsf{C}(\mathsf{A}) \to \mathsf{D}$ that inverts quasi-isomorphisms factors uniquely through $\mathsf{K}(\mathsf{A})$. This means that

1. The derived category factors uniquely through $\mathsf{K}(\mathsf{A})$;
2. Cohomology induces a functor $\mathsf{K}(\mathsf{A}) \to \mathsf{C}(\mathsf{A})$.

1.5.3 Complexes of Projective and Injective Objects

We will now focus on objects for which there is little difference between homotopy and quasi-isomorphism.

Definition 1.14: Projective/Injective

Let A be an abelian category. An object P of A is projective if $\mathrm{Hom}_{\mathsf{A}}(P, -)$ is exact, and an object I is injective if $\mathrm{Hom}_{\mathsf{A}}(-, I)$ is exact.

In arbitrary categories, it suffices to show a statement is true for only projectives or injectives to show it is true for the other, as projectives become injectives in A^{op}, and vice versa. It is only in specialized categories such as R-Mod where this symmetry may break down.

Definition 1.15: Enough Projectives/Injectives

A category has enough projectives if for every object $A \in \mathsf{A}$, there exists a projective object P and an epimorphism $P \twoheadrightarrow A$. A category has enough injectives if $\forall A \in \mathsf{A}$, there exists $I \in \mathsf{A}$ and a monomorphism $I \rightarrowtail A$.

A category does not in general have enough projectives and injectives, but some important ones do. For example, R-Mod with $R \in \mathsf{CRing}$, and the category of sheaves of abelian groups over a topological space all have enough projectives and injectives, the former of which we verified. However, counterexamples exist all over the place. For example, the abelian category of finite abelian groups has no nontrivial projectives and injectives, and 0 is not mono or epi.

1.5.4 Homotopy Equivalence vs. Quasi-Isomorphisms in $\mathsf{K}(\mathsf{A})$

We will herein prove that in $\mathsf{K}^-(\mathsf{P})$ and $\mathsf{K}^+(\mathsf{I})$ (where K^- denotes a bdd above complex), quasi-isomorphisms are isomorphisms. This will be done by showing that for any quasi-isomorphism $\alpha^\bullet : P_0^\bullet \to P_1^\bullet$ for $P_i^\bullet$ bounded above complexes of projectives, or likewise bounded below complexes of injectives, the morphism $\alpha^\bullet$ is a homotopy equivalence.

One remark I want to clarify is that if $A^\bullet$ is exact, then $H^\bullet(A^\bullet) = 0^\bullet$, and thus the identity map and 0 should be identified with each other in cohomology. However, there exist examples where these maps are not *homotopic*, so we need to specialize our assertion. First, I will rephrase some basic facts about projectives and injectives for general abelian categories. Note that since A^{op} is abelian, the following statements suffice to show the dual statement as well.

Lemma 1.15

The functor $\mathrm{Hom}_{\mathsf{A}}(X, -) : \mathsf{A} \to \mathsf{Ab}$ is left-exact. Additionally, an object P is projective if and only if for all objects B, C and for all epimorphisms $\psi : B \to C$, if $\eta : P \to C$ is a morphism, then there exists a morphism $\zeta : P \to B$ making the diagram

$$\begin{tikzcd} & P \arrow[dl, "\zeta"'] \arrow[d, "\eta"] & \\ B \arrow[r, two heads, "\psi"] & C \arrow[r, "0"] & 0 \end{tikzcd}$$

commute.

Proof. Because Ab has elements, this can be proven immediately by appealing to elements. Let

$$0 \longrightarrow A \xrightarrow{\phi} B \xrightarrow{\psi} C \longrightarrow 0$$

be exact in A. Then ϕ is mono, so $\ker\phi$ viewed as a morphism $\mathrm{Hom}_{\mathsf{A}}(X, A) \to \mathrm{Hom}_{\mathsf{A}}(X, B)$ is trivial, since the action on morphisms is given by $\eta \mapsto \phi\eta$, and this equals 0 iff $\eta = 0$. Note immediately that since $\psi\phi = 0$ in A, we must have $\psi\phi\eta = 0$ for all $\eta \in \mathrm{Hom}_{\mathsf{A}}(X, A)$, and thus $\mathrm{im}\,\phi \subseteq \ker\psi$. Now let $\zeta : X \to B$, and suppose $\psi\zeta = 0$. Then ζ must factor uniquely through $\mathrm{im}\,\phi$ since $\ker\psi = \mathrm{im}\,\phi$ in A. Since ϕ is mono, we have $\mathrm{im}\,\phi = \phi$, and thus the diagram

$$\begin{tikzcd} X \arrow[r, "\zeta"] \arrow[dr, "\bar\zeta"'] & B \arrow[r, "\psi"] & C \\ & A = \mathrm{im}\,\phi \arrow[u, "\phi"'] & \end{tikzcd}$$

commutes. Therefore, ζ is equivalent to a composition $\phi\bar\zeta$, and $\ker\psi \subseteq \mathrm{im}\,\phi$. Therefore, the complex

$$0 \longrightarrow \mathrm{Hom}_{\mathsf{A}}(X, A) \xrightarrow{\phi} \mathrm{Hom}_{\mathsf{A}}(X, B) \xrightarrow{\psi} \mathrm{Hom}_{\mathsf{A}}(X, C)$$

is exact.

Now suppose P is projective. It follows that

$$0 \longrightarrow \mathrm{Hom}_{\mathsf{A}}(P, A) \xrightarrow{\phi} \mathrm{Hom}_{\mathsf{A}}(P, B) \xrightarrow{\psi} \mathrm{Hom}_{\mathsf{A}}(P, C) \longrightarrow 0$$

is exact. It follows that the induced map ψ must be epi, and thus surjective. It follows that for all maps $\eta : P \to C$, there exists a map $\zeta : P \to B$ such that the diagram

$$\begin{tikzcd}
& P \arrow[d, "\eta"] \arrow[dr, "\zeta"] & \\
B \arrow[r, "\psi"] & C \arrow[r] & 0
\end{tikzcd}$$

commutes. ■

Lemma 1.16

Let $P^\bullet$ be a complex of projective objects with $P^i = 0$ for all $i > 0$, and let $L^\bullet$ be a complex satisfying $H^i(L^\bullet) = 0$ for all $i < 0$. Then for any morphism $\alpha^\bullet : P^\bullet \to L^\bullet$ which induces the 0 morphism in cohomology, it follows that α is homotopic to 0.

Proof. We need to find morphisms $h^i : P^i \to L^{i-1}$ satisfying

$$\alpha^i = d_L^{i-1} h^i + h^{i+1} d_P^i.$$

Note that $h^i = 0$ for $i > 0$, due to the bounds on $P^\bullet$. Note that since $L^1 = 0$, we must have $d_L^0 = 0$, and thus L^0 suffices as a kernel for d_L^{-1}. We thus have the diagram

$$\begin{tikzcd}
P^{-1} \arrow[r, "d_P^{-1}"] \arrow[d, "\alpha^{-1}"] & P^0 \arrow[r, "0"] \arrow[d, "\alpha^0"] & 0 \arrow[d, "0"] \\
L^{-1} \arrow[r, "d_L^{-1}"] \arrow[d, two heads, "\psi_L"] & L^0 \arrow[r, "0"] & 0 \\
\operatorname{Im} d_L^{-1} \arrow[ur, tail, "\operatorname{im} d_L^{-1}"] \arrow[r, tail, "\operatorname{im} d_L^{-1}"'] & \operatorname{Ker} d_L^0 = L^0 \arrow[u, "1_{L^0}"'] \arrow[r, two heads, "\operatorname{coker} \operatorname{im} d_L^{-1}"'] & H^0(L^\bullet)
\end{tikzcd}$$

We also have the diagram

$$\begin{tikzcd}
P^{-1} \arrow[r, two heads] \arrow[d, "\alpha^{-1}"] & \operatorname{Im} d_P^{-1} \arrow[r, tail] & \operatorname{Ker} d_P^0 = P^0 \arrow[r, two heads] \arrow[d, "\sigma = \alpha_0"] & H^0(P^\bullet) \arrow[d, "0"] \\
L^{-1} \arrow[r, two heads, "\psi_L"'] & \operatorname{Im} d_L^{-1} \arrow[r, tail, "\operatorname{im} d_L^{-1}"'] & \operatorname{Ker} d_L^0 = L^0 \arrow[r, two heads, "\operatorname{coker} \operatorname{im} d_L^{-1}"'] & H^0(L^\bullet)
\end{tikzcd}$$

Since $\operatorname{Im} d_L^{-1} \to \operatorname{Ker} d_L^0$ is mono, by commutivity of the diagram (in particular, the right square), We find that $\operatorname{coker} \operatorname{im} d_L^{-1} \sigma = 0$, and thus σ factors through $\ker \operatorname{coker} \operatorname{im} d_L^{-1}$, which is simply $\operatorname{im} d_L^{-1}$ since images are mono. Therefore, there is an induced map $P^0 \to \operatorname{Im} d_L^{-1}$ that makes the diagram commute. Since there exists an epimorphism $\psi_L : L^{-1} \to \operatorname{Im} d_L^{-1}$, by lemma 1.15 the map $P^0 \to \operatorname{Im} d_L^{-1}$ raises to a map

$h^0 : P^0 \to L^{-1}$ in such a way that the diagram

$$\begin{tikzcd}
& & P^0 \arrow[dll, "h^0"'] \arrow[d, "\alpha_0"] \\
L^{-1} \arrow[r, "\psi_L"'] \arrow[rr, bend right, "d_L^{-1}"'] & \operatorname{Im} d_L^{-1} \arrow[r, "\operatorname{im} d_L^{-1}"] & L^0
\end{tikzcd}$$

commutes. Therefore, $\alpha_0 = d_L^{-1} h^0$, and since $h^1 = 0$, this satisfies the homotopy condition.

We continue inductively. Suppose h^i and h^{i+1} exist. Then we must have

$$\alpha^i = d_L^{i-1} h^i + h^{i+1} d_P^i \implies \alpha^i d_P^{i-1} = d_L^{i-1} h^i d_P^{i-1}.$$

It follows that

$$d_L^{i-1}\left(\alpha^{i-1} - h^i d_P^{i-1}\right) = d_L^{i-1}\alpha^{i-1} - \alpha^i d_P^{i-1}.$$

Since α is a morphism of chain complexes, we must have $d_L^{i-1}\alpha^{i-1} = \alpha^i d_P^{i-1}$, and thus

$$d_L^{i-1}\left(\alpha^{i-1} - h^i d_P^{i-1}\right) = 0.$$

Therefore, $\alpha^{i-1} - h^i d_P^{i-1}$ factors through $\ker d_L^{i-1}$. Since $L^\bullet$ is exact here, since we have already handled all cases where it isn't exact, we have $\operatorname{Im} d_L^{i-2} = \operatorname{Ker} d_L^{i-1}$, and thus the diagram

$$\begin{tikzcd}
\operatorname{Im} d_L^{i-2} = \operatorname{Ker} d_L^{i-1} \arrow[dr, hook, "\ker d_L^{i-1}", "\operatorname{im} d_L^{i-2}"'] & P^{i-1} \arrow[l, dashed] \arrow[d, "\alpha^{i-1} - h^i d_P^{i-1}"] \arrow[dr, bend left, "0"] & \\
L^{i-2} \arrow[u, dashed, two heads] \arrow[r, "d_L^{i-2}"'] & L^{i-1} \arrow[r, "d_L^{i-1}"'] & L^i
\end{tikzcd}$$

commutes. By lemma 1.15, since $L^{i-2} \to \operatorname{Im} d_L^{i-2}$ is epi, the morphism $P^{i-1} \to \operatorname{Ker} d_L^{i-1}$ raises to a morphism $h^{i-1} : P^{i-1} \to L^{i-2}$ such that

$$\begin{tikzcd}
& P^{i-1} \arrow[dl, "h^{i-1}"'] \arrow[d, "\alpha^{i-1} - h^i d_P^{i-1}"] \\
L^{i-2} \arrow[r, "d_L^{i-2}"'] & L^{i-1}
\end{tikzcd}$$

commutes, where we use commutivity of the above diagram to justify the morphism d_L^{i-2}. Therefore,

$$\alpha^{i-1} - h^i d_P^{i-1} = d_L^{i-2} h^{i-1} \implies \alpha^{i-1} = d_L^{i-2} h^{-i1} + h^i d_P^{i-1}.$$

This concludes the proof. ■

Corollary 1.3

If $P^\bullet$ is a bounded above complex of projectives, and $L^\bullet$ is exact, then any morphism $P^\bullet \to L^\bullet$ is homotopic to 0.

The dual is that for any complex in $\mathsf{C}^+(\mathsf{I})$, all morphisms into the object are homotopic to 0. Moreover, any bounded above exact complex of projectives is easily shown to be homotopy equivalent to 0.

Corollary 1.4

Let $P^\bullet$ be a bounded above exact chain complex of projectives. Then $P^\bullet$ is homotopy equivalent to 0.

Lemma 1.17

Let $\rho^\bullet : L^\bullet \to M^\bullet$ be a quasi-isomorphism in $\mathsf{C}(\mathsf{A})$, and let $P^\bullet$ be a bounded above complex of projectives. Let $\alpha^\bullet : P^\bullet \to L^\bullet$ be a morphism such that $\rho^\bullet \circ \alpha^\bullet \sim 0$. Then $\alpha^\bullet \sim 0$.

Proof. Let $h^\bullet$ give the homotopy $\rho^\bullet \circ \alpha^\bullet \sim 0$. Define it such that

$$-\rho^i \alpha^i = d_M^{i-1} h^i + h^{i+1} d_P^i,$$

which is easily shown to be equivalent. Define $\beta^i : P^i \to L^i \oplus M^{i-1} = MC(\rho)^{i-1}$ as the map satisfying

$$\begin{tikzcd}
 & & L^i \\
P^i \arrow[r, dashed, "\beta^i"] \arrow[urr, bend left, "\alpha^i"] \arrow[drr, bend right, "h^i"'] & MC(\rho)^{i-1} \arrow[ur, "\pi_{L^i}"] \arrow[dr, "\pi_{M^{i-1}}"] & \\
 & & M^{i-1}
\end{tikzcd}$$

The claim is that $\beta^\bullet$ is a cochain morphism $P^\bullet \to MC(\rho)[-1]^\bullet$, where we use the nonstandard maps $-d^\bullet_{MC(\rho)}$ for the morphisms in $MC(\rho)^\bullet$. The proof of this is really painful to do arrow theoretically and I refuse to copy down a proof that relies on Freyd-Mitchell. However, it is fairly intuitive why it works.

Since $\rho^\bullet$ is a quasi-isomorphism, $MC(\rho)^\bullet$ is exact, and thus $\beta^\bullet \sim 0$. Since $\alpha^\bullet = \pi_L^\bullet \circ \beta^\bullet$, we find that $\alpha^\bullet \sim 0$. ■

1.5.5 Proof of Theorem 1.4

Proposition 1.5

Let A be abelian, and let $L^\bullet$ be a commplex in $\mathsf{C}(\mathsf{A})$. Let $P^\bullet$ in $\mathsf{C}^-(\mathsf{P})$, and let $\alpha^\bullet : L^\bullet \to P^\bullet$ be a quasi-isomorphism. Then there exists a morphism $\beta^\bullet : P^\bullet \to L^\bullet$ such that $\alpha^\bullet \circ \beta^\bullet \sim 1_{P^\bullet}$.

Proof. Since $\alpha^\bullet$ is a quasi-isomorphism, its mapping cone is exact. Let $\rho^\bullet : P^\bullet \to L[1]^\bullet \oplus M^\bullet = MC(\alpha)^\bullet$ be the morphism $0 \times 1_{P^\bullet}$. Since the mapping cone is exact, and $P^\bullet$ is bounded above, by corollary 1.4 $\rho^\bullet$ is homotopic to 0. Honestly the rest of the proof is trivial – you just unravel the induced homotopic maps. ■

Theorem 1.4

Let A be an abelian category, and let $\alpha^\bullet : P_0^\bullet \to P_1^\bullet$ be a quasi-isomorphism between elements of $\mathsf{C}^-(\mathsf{P})$ (bounded above complexes in P). Then $\alpha^\bullet$ is a homotopy equivalence.

Proof. By proposition 1.5, there exists a morphism $\beta^\bullet : P_1^\bullet \to P_0^\bullet$ such that $\alpha^\bullet \circ \beta^\bullet \sim 1_{P_1^\bullet}$. It follows that $H^\bullet(\alpha^\bullet) \circ H^\bullet(\beta^\bullet) = 1$. Therefore, $\beta^\bullet$ is a quasi-isomorphism, and thus by the proposition there again exists a morphism $\alpha^{\bullet\prime} : P_0^\bullet \to P_1^\bullet$ such that $\beta^\bullet\alpha^\bullet \sim 1_{P_0^\bullet}$. Since homotopy is an equivalence relation, it follows that $\beta^\bullet\alpha^\bullet \sim 1$, and $\alpha^\bullet$ is a homotopy equivalence. ■

1.6 Projective and Injective Resolutions and the Derived Category

Theorem 1.4 is very important - it tells us that if we are willing to restrict our attention to complexes of projectives that are bounded above, or complexes of injectives that are bounded below, then quasi-isomorphisms become precisely homotopy equivalences. This is important, since homotopic maps are more well-behaved than quasi-isomorphisms in general. For example, any additive functor that inverts quasi-isomorphisms preserves homotopy, whereas quasi-isomorphisms are not. Also, homotopy and homotopy equivalence are equivalence relations, while quasi-isomorphism is not. We can apply this to study more general objects of A using projective and injective resolutions. This can only really be done if A has enough projectives/injectives.

1.6.1 Recovering A

Recall that A can be embedded in $\mathsf{C}(\mathsf{A})$ as the subcategory $\iota(\mathsf{A})$. Composing with the inclusion $\mathsf{C}(\mathsf{A}) \to \mathsf{K}(\mathsf{A})$ gives us an embedding of A into the homotopic category. But there's more copies of A in $\mathsf{K}(\mathsf{A})$, which offer more insight.

Definition 1.16: Resolution

A projective resolution of A is a quasi-isomorphism $P^\bullet \to \iota(A)$ for $P^\bullet \in \mathsf{C}^{\leq 0}(\mathsf{P})$. An injective resolution of A is a quasi-isomorphism $\iota(A) \to Q^\bullet$, for $Q^\bullet \in \mathsf{C}^{\geq 0}(\mathsf{I})$.

Clearly, resolutions exist if a category has enough projectives/injectives, by the lemma characterizing them. Now consider the category of homotopy classes of quasi-isomorphisms with target $\iota(A)$. If projective resolutions exist, we will show they are initial in this category. Respectively, injective resolutions are final the category of homotopy classes of quasi-isomorphisms with source $\iota(A)$. This will show that in $\mathsf{K}(\mathsf{A})$, projective resolutions will map uniquely to *every* resolution of A.

Lemma 1.18

Let A be an object in an abelian category, and let $M^\bullet$ be a resolution of A. Let $P^\bullet \in \mathsf{C}^{\leq 0}(\mathsf{P})$, and let

$$\varphi : H^0(P^\bullet) \to H^0(M^\bullet) = A.$$

Then

- There exist morphisms $\alpha^\bullet : P^\bullet \to M^\bullet$ that induce φ in cohomology on the 0th degree;
- Different morphisms satisfying this requirement are homotopy equivalent.

Proof. First, we define $\alpha^\bullet$. Since $P^i = 0$ for $i > 0$, we must have $\alpha^i = 0$ for $i > 0$. It then suffices without loss of generality to consider the truncated version of $M^\bullet$. First, note that for any complex $A^\bullet$, there exists a projection $A^i \to H^i(A^\bullet)$, defined as follows:

$$\begin{tikzcd}
A^{i-1} \arrow[r, "d_A^{i-1}"] \arrow[d, two heads] & P^i \arrow[r, "d_A^i"] \arrow[d, dashed, "\pi"] & A^{i+1} \\
\operatorname{Im} d_A^{i-1} \arrow[r, "\phi"] \arrow[ur, tail, "\operatorname{im} d_A^{i-1}"] & \operatorname{Ker} d_A^i \arrow[r, "\operatorname{coker}\phi"] \arrow[u, bend left, "\ker d_A^i"] \arrow[ur, "0"] & H^i(A^\bullet)
\end{tikzcd}$$

where π exists by the universal property of kernels, and we define the projection as $\operatorname{coker}\phi \circ \pi$. We denote it simply by π.

We then have the diagram

$$\begin{tikzcd}
P^{-2} \arrow[r, "d_P^{-2}"] & P^{-1} \arrow[r, "d_P^{-1}"] & P^0 \arrow[r, "\pi"] & H^0(P^\bullet) \arrow[r] \arrow[d, "\varphi"] & 0 \\
M^{-2} \arrow[r, "d_M^{-2}"'] & M^{-1} \arrow[r, "\phi\psi"'] & \operatorname{Ker} d_M^0 \arrow[r, "\mu"'] & H^0(M^\bullet) \arrow[r] & 0
\end{tikzcd}$$

where the morphisms on the bottom complex are given by

$$\begin{tikzcd}
M^{-1} \arrow[r, "d_M^{-1}"] \arrow[d, two heads, "\psi"'] & M^0 & \\
\operatorname{Im} d_M^{-1} \arrow[r, "\phi"'] \arrow[ur, tail, "\operatorname{im} d_M^{-1}"] & \operatorname{Ker} d_M^0 \arrow[r, "\mu"] \arrow[u, tail] & \operatorname{Coker}\phi = H^0(M^\bullet)
\end{tikzcd}$$

The bottom complex is exact since $M^\bullet$ is exact except at M^0, but we have replaced M^0 with $\ker d_M^0$, after doing a short diagram chase. Since P^0 is projective, φ lifts to a morphism

$$\alpha^0 : P^0 \longrightarrow \operatorname{Ker} d_M^0 \rightarrowtail M^0.$$

Further note that by commutivity of the diagram in question,

$$\mu\alpha^0 d_P{-}1 = \varphi\pi d_P^{-1} = 0$$

since rows are complexes. Therefore, $\alpha^0 d_P^{-1}$ factors through $\ker\mu = \operatorname{im} d_M^{-1}$, and thus we have a diagram

$$\begin{tikzcd}
 & P^{-1} \arrow[d, "\alpha^0 d_P^{-1}"] \\
M^{-1} \arrow[r, two heads] & \operatorname{Im} d_M^{-1}
\end{tikzcd}$$

which lifts to a morphism $\alpha^{-1} : P^{-1} \to M^{-1}$. This generalizes inductively, and the construction ensures commutivity of the relevant diagrams. Therefore, $\alpha^\bullet : P^\bullet \to M^\bullet$ exists, and uniqueness up to homotopy follows as a corollary of lemma 9.16. ∎

An important, albeit immediate, corollary of this result is as follows:

Proposition 1.6

Any projective/injective resolutions of an object are homotopy equivalent.

The idea behind this lemma is actually fairly strong - for any object A of A, we can associate an object $\mathsf{K}^{\leq 0}(\mathsf{P})$ to A (provided enough projectives exist), determined up to homotopy. This is in fact functorial.

Proposition 1.7

Let A_0, A_1 be objects of an abelian category, and let $P_i^\bullet$ be a projective resolution of A_i. Then every morphism $\varphi : A_0 \to A_1$ is induced by a morphism $\alpha^\bullet : P_0^\bullet \to P_1^\bullet$, which is determined up to homotopy.

Proof. By hypothesis, $\varphi : H^0(P_0^\bullet) \to H_0(P_1^\bullet)$. By lemma 1.18, there exists $\alpha^\bullet$ inducing the morphism φ, and it is unique up to homotopy. ■

1.6.2 From Objects to Complexes

This assignment is clearly covariant, as shown by previous proofs. This shows that objects with projective resolutions in A can be viewed as elements of $\mathsf{K}^-(\mathsf{P})$, and vice versa for injectives. This forms a full subcategory of A in the respective homotopic category, provided that A has enough injectives. It is also determined up to homotopy equivalence, which is inverted in the homotopy category.

We can thus treat each object A as its isomorphism class in $\mathsf{K}^-(\mathsf{P})$ of projective resolutions, and morphisms in A correspond to morphisms between projective resolutions. This preserves the structure of A, and we can in fact sort of replace A with $\mathsf{K}^-(\mathsf{P})$.

Since homotopy equivalent complexes have the same cohomology, and additive functors preserve homotopy equivalence, objects in this category carry cohomological information in the optimal way. We would think that $\mathsf{K}^-(\mathsf{P})$ and $\mathsf{K}^+(\mathsf{I})$ should satisfy the criterion for a derived category, if we are only considering something like $\mathsf{C}^+(\mathsf{A})$ for A having enough projectives. The following theorem will finalize this notion as far as we can in this book.

Theorem 1.5

Let A be an abelian category with enough projectives, and let $L^\bullet$ be a complex in $\mathsf{C}^-(\mathsf{A})$. Then there exists a bounded above complex of projectives $P^\bullet$ and a quasi-isomorphism $P^\bullet \to L^\bullet$, and $P^\bullet$ is uniquely defined up to homotopy equivalence (isomorphism in the homotopic category). Further, every morphism $\alpha^\bullet$ of complexes in $\mathsf{C}^-(\mathsf{A})$ lifts to a corresponding morphism of projective resolutions in $\mathsf{K}^-(\mathsf{P})$, also uniquely determined up to homotopy.

Proof. First, we construct $P^\bullet$. This will be done while inductively constructing $\lambda^\bullet : P^\bullet \to L^\bullet$, and will follows from the fact that A has enough projectives. Recall the explicit definition of a pullback, which exists in abelian categories: for any diagram

$$\begin{tikzcd} & X \arrow[d, "f"] \\ Y \arrow[r, "g"'] & Z \end{tikzcd}$$

in an abelian category, there exists a diagram

$$\begin{tikzcd}
X \times Y \arrow[r, "\pi_X"] \arrow[d, "\pi_Y"'] & X \arrow[d, "f"] \\
Y \arrow[r, "g"'] & Z
\end{tikzcd}$$

The kernel P of the morphism $h \coloneqq f \circ \pi_X - g\pi_Y : X \times Y \to Z$, together with the morphisms $p_X \coloneqq \pi_X \circ \ker h$, $p_Y \coloneqq \pi_Y \circ \ker h$ satisfies the universal property for pullbacks

$$\begin{tikzcd}
A \arrow[dr, dashed] \arrow[drr, bend left] \arrow[ddr, bend right] & & \\
& P \arrow[r, "p_X"] \arrow[d, "p_Y"'] & X \arrow[d, "f"] \\
& Y \arrow[r, "g"'] & Z
\end{tikzcd}$$

Suppose inductively that $\lambda^{i+1} : P^{i+1} \to L^{i+1}$ exists. Then there exists a pullback

$$\begin{tikzcd}
L^i \times_{\operatorname{Ker} d_L^{i+1}} \operatorname{Ker} d_P^{i+1} \arrow[r, "\pi"] \arrow[d, two heads] & \operatorname{Ker} d_P^{i+1} \arrow[d, two heads, "\hat{\lambda}^{i+1}"] \\
L^i \arrow[r, "\overline{d}_L^i"'] & \operatorname{Ker} d_L^{i+1}
\end{tikzcd}$$

The morphism $\hat{\lambda}^{i+1}$ is epi by inductive hypothesis, and thus so is the projection from the pullback to L^i. The morphism $\overline{d}_L^i$ is the restriction of the target of d_L^i – in my notation, it would be $\phi_L^i \psi_L^i$.

Since A has enough projectives, there exists a projective P^i and an epimorphism

$$P^i \twoheadrightarrow L^i \times_{\operatorname{Ker} d_L^{i+1}} \operatorname{Ker} d_P^{i+1}$$

We can then immediately define d_P^i as the composition

$$\begin{tikzcd}
P^i \arrow[r, two heads] \arrow[rrr, bend left, "d_P^i"] & L^i \times_{\operatorname{Ker} d_L^{i+1}} \operatorname{Ker} d_P^{i+1} \arrow[r, "\pi"'] & \operatorname{Ker} d_P^{i+1} \arrow[r, tail, "\ker d_P^{i+1}"'] & P^{i+1},
\end{tikzcd}$$

which trivially makes $P^\bullet$ into a complex inductively. By lemma 1.15, since there exists an epimorphism from the pullback to L^i, and there exists a map from P^i into the pullback, this lifts into a map λ^i making the diagram

$$\begin{tikzcd}
P^i \arrow[r, two heads] \arrow[rrr, bend left, "d_P^i"] \arrow[dr, two heads, "\lambda^i"'] & L^i \times_{\operatorname{Ker} d_L^{i+1}} \operatorname{Ker} d_P^{i+1} \arrow[r, "\pi"'] \arrow[d, two heads] & \operatorname{Ker} d_P^{i+1} \arrow[r, tail, "\ker d_P^{i+1}"'] \arrow[d, two heads, "\hat{\lambda}^{i+1}"] & P^{i+1} \\
& L^i \arrow[r, "\overline{d}_L^i"'] & \operatorname{Ker} d_L^{i+1} &
\end{tikzcd}$$

commute. We find that λ^i is epi since it is a composition of epimorphisms. The final thing to prove is that the induced map $\hat{\lambda}^i$ is epi, which follows immediately.

To show that $\lambda^\bullet$ is inverted by cohomology, Note that $H^{i+1}(P^\bullet)$ and $H^{i+1}(L^\bullet)$ can be realized as the cokernels of π and $\overline{d}_L^i$, respectively. Since $\hat{\lambda}^{i+1}$ is epi, we must then have the map $\overline{d}_L^i \oplus -\hat{\lambda}^{i+1}$ is epi, and thus we find that the diagram is a pushout as well, by a previous exercise. This induces the isomorphism between cohomologies, by a previous exercise.

Let $P^\bullet, \overline{P}^\bullet$ be bounded complexes of projectives, and let $\lambda^\bullet, \overline{\lambda}^\bullet$ be the corresponding quasi-isomorphisms into $L^\bullet$. We want to find a homotopy equivalence $\widetilde{\lambda}^\bullet : \overline{P}^\bullet \to P^\bullet$.

First, we construct such a morphism. Note tht $\widetilde{\lambda}^i = 0$ for $i > 0$, so assume the inductive hypothesis. We have a diagram

$$\begin{tikzcd}
P^i \arrow[r, two heads] \arrow[dr, two heads, "\lambda^i"'] & L^i \times_{\operatorname{Ker} d_L^{i+1}} \operatorname{Ker} d_M^{i+1} \arrow[r, "\pi"] \arrow[d, two heads] & \operatorname{Ker} d_P^{i+1} \arrow[r, tail, "\ker d_P^{i+1}"] \arrow[d, two heads, "\hat{\lambda}^{i+1}"] & P^{i+1} \arrow[d, two heads, "\lambda^{i+1}"] \\
& L^i \arrow[r, "\overline{d}_L^i"] & \operatorname{Ker} d_L^{i+1} \arrow[r, tail, "\ker d_L^{i+1}"] & L^{i+1} \\
\overline{P}^i \arrow[ur, "\overline{\lambda}^i"] \arrow[rrr, "d_{\overline{P}}^i"] & & & \overline{P}^{i+1} \arrow[u, "\overline{\lambda}^{i+1}"']
\end{tikzcd}$$

Note that by inductive hypothesis, the diagram

$$\begin{tikzcd}
P^i \arrow[r] & P^{i+1} \arrow[r] & P^{i+2} \\
\overline{P}^i \arrow[r] \arrow[rr, bend right, "0"'] & \overline{P}^{i+1} \arrow[r] \arrow[u, "\lambda^{i+1}"] & \overline{P}^{i+2} \arrow[u, "\lambda^{i+2}"]
\end{tikzcd}$$

commutes, and thus the composition

$$\begin{tikzcd}
\overline{P}^i \arrow[r, "d_{\overline{P}}^i"'] \arrow[rrr, bend left, "0"] & \overline{P}^{i+1} \arrow[r, "\lambda^{i+1}"'] & P^{i+1} \arrow[r, "d_P^{i+1}"'] & P^{i+2}
\end{tikzcd}$$

is 0. Therefore, it factors unqiuely through the kernel:

$$\begin{tikzcd}
\overline{P}^i \arrow[r, "d^i_{\overline{P}}"] \arrow[drr, dashed, "\varepsilon"'] & \overline{P}^{i+1} \arrow[r, "\lambda^{i+1}"] & P^{i+1} \arrow[r, "d_P^{i+1}"] & P^{i+2} \\
& & \operatorname{Ker} d_P^{i+1} \arrow[u, tail, "d_P^{i+1}"] &
\end{tikzcd}$$

By commutivity of the main diagram, this gives us the commutative square

$$\begin{tikzcd}
\overline{P}^i \arrow[r, "\varepsilon"] \arrow[d, "\overline{\lambda}^i"'] & \operatorname{Ker} d_M^{i+1} \arrow[d, two heads, "\hat{\lambda}^{i+1}"] \\
L^i \arrow[r, "\overline{d}^i_L"'] & \operatorname{Ker} d_L^{i+1}
\end{tikzcd}$$

Since the pullback is final with respect to these diagrams, there is an induced morphism making the diagram

$$\begin{tikzcd}
\overline{P}^i \arrow[dr, dashed, "\sigma"] \arrow[drr, bend left, "\varepsilon"] \arrow[ddr, bend right, "\overline{\lambda}^i"'] & & \\
& L^i \times_{\operatorname{Ker} d_L^{i+1}} \operatorname{Ker} d_M^{i+1} \arrow[r] \arrow[d, two heads] & \operatorname{Ker} d_M^{i+1} \arrow[d, two heads, "\hat{\lambda}^{i+1}"] \\
& L^i \arrow[r, "\overline{d}^i_L"'] & \operatorname{Ker} d_L^{i+1}
\end{tikzcd}$$

commute. Returning to the main diagram, we now have morphisms making the diagram

$$\begin{tikzcd}
P^i \arrow[r, two heads] \arrow[dr, "\lambda^i"] & L^i \times_{\operatorname{Ker} d_L^{i+1}} \operatorname{Ker} d_M^{i+1} \arrow[r, "\pi"] \arrow[d, two heads] & \operatorname{Ker} d_P^{i+1} \arrow[r, tail, "\ker d_P^{i+1}"] \arrow[d, two heads, "\hat{\lambda}^{i+1}"] & P^{i+1} \arrow[d, two heads, "\lambda^{i+1}"] \\
& L^i \arrow[r, "\overline{d}^i_L"] & \operatorname{Ker} d_L^{i+1} \arrow[r, tail, "\ker d_L^{i+1}"] & L^{i+1} \\
\overline{P}^i \arrow[uur, bend left, "\sigma"] \arrow[ur, "\overline{\lambda}^i"] \arrow[uurr, bend right, "\varepsilon"] \arrow[rrr, "d^i_{\overline{P}}"] & & & \overline{P}^{i+1} \arrow[u, "\overline{\lambda}^{i+1}"']
\end{tikzcd}$$

commute. Since $\overline{P}^i$ is projective, there is an induced morphism $\widetilde{\lambda}^i$ making the diagram

$$\begin{tikzcd}
& \overline{P}^i \arrow[dl, "\widetilde{\lambda}^i"'] \arrow[d, "\sigma"] \\
P^i \arrow[r, two heads] & L^i \times_{\operatorname{Ker} d_L^{i+1}} \operatorname{Ker} d_M^{i+1}
\end{tikzcd}$$

commute. We thus have a diagram in $\mathsf{C}^-(\mathsf{A})$:

$$\begin{tikzcd} & P^\bullet \arrow[d, two heads, "\lambda^\bullet"] \\ \overline{P}^\bullet \arrow[ur, "\tilde{\lambda}^\bullet"] \arrow[r, "\overline{\lambda}^\bullet"'] & L^\bullet \end{tikzcd}$$

It follows that $\widetilde{\lambda}^\bullet$ is a quasi-isomorphism, since the other two are. By theorem 1.4, it must be a homotopy equivalence. Therefore, $P^\bullet$ is unique up to homotopy equivalence.

By this argument, we also have that for any morphism $\alpha^\bullet : \overline{P}^\bullet \to L^\bullet$, for $\overline{P}^\bullet \in \mathsf{C}^-(\mathsf{P})$, then this lifts to a morphism $\overline{P}^\bullet \to P^\bullet$ making the relevant diagram commute. In fact, for any $\overline{P}$ mapping quasi-isomorphically to $L^\bullet$, the morphism lifts up to homotopy to $P^\bullet$, since it must be homotopic to the previously constructed one. So suppose morphisms $\alpha_0^\bullet$ and $\alpha_1^\bullet$ lift $\alpha^\bullet$ up to homotopy:

$$\begin{tikzcd} \overline{P}^\bullet \arrow[r, shift left, "\alpha_0^\bullet"] \arrow[r, shift right, "\alpha_1^\bullet"'] \arrow[d] & P^\bullet \arrow[d, "\lambda^\bullet"] \\ \overline{L}^\bullet \arrow[r, "\alpha^\bullet"'] & L^\bullet \end{tikzcd}$$

Then $\lambda^\bullet(\alpha_1^\bullet - \alpha_0^\bullet) \sim 0$, and thus $\alpha_1^\bullet - \alpha_0^\bullet \sim 0$ by a previous lemma. Therefore, $\alpha_1^\bullet \sim \alpha_0^\bullet$. ■

1.6.3 Poor Man's Derived Category

We can now close out on the important ideas we've been working towards. If A has enough projectives, then we can view $\mathsf{K}^-(\mathsf{P})$ as a partial solution to the universal problem of the derived category.

Theorem 1.6

Let A be a category with enough projectives. Then by lemma 1.13, the functor $\mathsf{C}^-(\mathsf{A}) \to \mathsf{D}^-(\mathsf{A})$ factors through $\mathsf{K}^-(\mathsf{A}) \to \mathsf{D}^-(\mathsf{A})$. This induces an equivalence of categories $\mathsf{K}^-(\mathsf{P}) \to \mathsf{D}^-(\mathsf{A})$. Similarly, if A has enough injective, then $\mathsf{K}^+(\mathsf{I}) \to \mathsf{D}^+(\mathsf{A})$ is an equivalence of categories.

We cannot prove this since we have not precisely defined the derived category, nor have we defined methods of proving universal problems between categories. However, we can appreciate *why* it should be true.

Suppose A has enough projectives. Let $\mathscr{P} : \mathsf{C}^-(\mathsf{A}) \to \mathsf{K}^-(\mathsf{A})$ be the functor mapping $L^\bullet$ to some projective resolution, which exists by theorem 1.5, with every morphism $\alpha^\bullet : L^\bullet \to M^\bullet$ mapping to the morphism in $\mathsf{K}^-(\mathsf{P})$ that lifts $\alpha^\bullet$.

Since we have shown any two lifts are homotopic, $\mathscr{P}(\alpha^\bullet)$ is well-defined and is thus easily shown to be a functor. Additionally, since $P^\bullet$ is well-defined up to homotopy equivalence, we have that each $L^\bullet$ maps to some element of an isomorphism class, and is well-defined up to isomorphism.

This can be rephrased by saying that any two choices of functor $\mathscr{P}_1, \mathscr{P}_2$ are naturally isomorphic; there exists a natural isomorphism $\eta : \mathscr{P}_1 \Rightarrow \mathscr{P}_2$.

Choose $\mathscr{P}$. The claim is that this functor solves the universal problem. The following theorem is our best way of stating this without going more into the theory.

Theorem 1.7

With notation as above, we have the following.

- If $\rho\circ$ is a quasi-isomorphism in $\mathsf{C}^-(\mathsf{A})$, then $\mathscr{P}(\rho^\bullet)$ is an isomorphism in $\mathsf{K}^-(\mathsf{P})$.
- Let $\mathscr{F} : \mathsf{C}^-(\mathsf{A}) \to \mathsf{D}$ be an additive functor that inverts quasi-isomorphisms. Then there exists $\widetilde{\mathscr{F}} : \mathsf{K}^-(\mathsf{P}) \to \mathsf{D}$ making the diagram

$$\begin{tikzcd} \mathsf{C}^-(\mathsf{A}) \arrow[r, "\mathscr{F}"] \arrow[d, "\mathscr{P}"'] & \mathsf{D} \\ \mathsf{K}^-(\mathsf{P}) \arrow[ur, "\widetilde{\mathscr{F}}"'] & \end{tikzcd}$$

  commute up to isomorphism. That is, there exists a natural transformation $\nu : \widetilde{\mathscr{F}} \circ \mathscr{P} \Rightarrow \mathscr{F}$ such that

$$\nu_{L^\bullet} : \widetilde{\mathscr{F}} \circ \mathscr{P}(L^\bullet) \to \mathscr{F}(L^\bullet)$$

  is an isomorphism for all $L^\bullet$. Moreover, $\widetilde{\mathscr{F}}$ is unique up to natural isomorphism.

The noncommutivity of the diagram (not up to isomorphism) is due to the fact that this is “only” an equivalence of categories, not a categorical isomorphism.

Proof. a ■